BFGS on Convex Nonsmooth Functions: Convergence and Failure

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Abstract Ordinary full BFGS often succeeds on nonsmooth objectives, but its convergence theory must accommodate gradient jumps and possible degeneration of the quasi-Newton matrices. We give complementary positive and negative results for finite convex objectives with Armijo–weak-Wolfe steps and actual gradients at differentiable accepted iterates. A centered smooth–sublinear theorem proves global whole-sequence convergence for $u^2 + |v|$ and applies in arbitrary dimension on compact initial sublevels. Coercive degree-one homogeneous objectives, including the stated max-linear and gauge families, admit global convergence. Local results cover finite polyhedral smooth-piece partitions and nondegenerate planar smooth maxima; an explicit curved-interface example requires moving comparison geometry beyond the radial criterion. In contrast, a fixed unit-first implementation can converge to nonstationary points from starts arbitrarily close to the unique minimizer of one fixed objective. One construction is a maximum of four globally convex $C^{2,1}$ branches; another retains uniform strong convexity with $C^{1,1}$ branches that lose second differentiability at the accumulation point. The positive proofs derive an adjacent-degree hierarchy for actual principal minors of full-BFGS matrices, then combine reciprocal growth with comparison energies. The resulting objective-level conditions require neither iterate convergence nor prescribed spectral behavior. The negative proofs certify the complete trial trees and infinite local embeddings. These results expand the convergence theory while ruling out unrestricted deterministic guarantees for the specified classes.

Keywords BFGS · convex nonsmooth optimization · global convergence · local convergence · counterexample

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1 Introduction

BFGS builds its search directions from changes in gradients, yet it often works well when those gradients jump. Classical convex convergence theory uses smoothness together with the evolving quasi-Newton matrices [9, 8, 7]. Lewis–Overton's experiments and analyses [1] exposed the difficulty of extending that theory: nonsmooth BFGS can approach a minimizer while its inverse-Hessian approximations become highly singular. What global or local convergence can

ordinary full BFGS guarantee on convex nonsmooth functions, and how can one prove it without assuming the asymptotic behavior of its iterates or matrices?

We give three answers: deterministic counterexamples to every-start local convergence; global and local convergence theorems for structured convex families, including the mixed benchmark below; and a principal-minor and comparison-energy argument that remains effective as the BFGS matrices degenerate.

Lewis–Overton's Challenge 7.1 [1] gives this question a broader, probabilistic setting. For locally Lipschitz semialgebraic objectives with bounded sublevels and absolutely continuously distributed initial data, it asks about nonbreakdown, stationary cluster points, R-linear convergence of all function trial values, and identification of spectral subspaces under partly smooth convergence. We study deterministic guarantees for convex objectives. Successful termination is allowed; infinite runs are analyzed under the differentiable-iterate convention made precise below.

Substantial positive theory precedes this work. Guo–Lewis extends Powell's argument through smooth convex replacements on regions relevant to a BFGS sequence [5, Theorems 3.2, 4.1 and 4.3]. The resulting theory proves Euclidean-norm convergence in arbitrary finite dimension [5, Corollary 4.2]; for the mixed model

$$F(u, v) = u^2 + |v|$$

Corollary 4.4 gives only a subsequence approaching a point of $v = 0$. The following paragraph explicitly leaves convergence of every BFGS sequence to $(0, 0)$ open [5, Corollary 4.4 and following discussion]. Approaching the nonsmooth line does not force the smooth coordinate u to vanish. Corollary 2 answers this whole-sequence question for every admissible full-BFGS run, as a consequence of an arbitrary-dimensional family theorem rather than a special planar recurrence.

Two other advances clarify the remaining task. Xie–Wächter's two-piece unbounded affine model leads to an unbounded descent direction after finitely many iterations (Xie and Wächter, 2017 preprint, arXiv:1712.08571v1, Theorem IV.1); a value-convergence theorem for a broader homogeneous family does not replace that finite escape result. Gebken's published 2026 introduction still identifies $u^2 + |v|$ as unresolved. His local criticality theorem for continuous smooth selections assumes an iterate limit, repeated activity, and specified spectral and gradient-jump behavior [3, Theorem 3.1 and (B1)]. Our positive results instead derive convergence from the objective hypotheses for the families stated here.

Throughout the paper, an *admissible run* uses the unmodified full BFGS update, an arbitrary $H_0 \succ 0$, fixed $0 < c_1 < c_2 < 1$, Armijo–weak-Wolfe steps, and differentiable accepted points with nonzero gradients (Definition 1). The positive theorems concern every infinite such run. They do not assert that every search ray contains an acceptable differentiable point.

Deterministic limits. We construct convex objectives for which a fixed unit-first Armijo–weak-Wolfe doubling/bisection search generates full-BFGS iterates

converging to a nonstationary point. For each local construction, one fixed objective has failing starts arbitrarily close to its unique minimizer, all with $H_0 = I$, and every accepted or rejected query approaches that minimizer across the family of runs. One construction is a maximum of four convex $C^{2,1}$ branches with bounded Hessians; another is a maximum of four uniformly strongly convex $C^{1,1}$ branches, three of which are not twice differentiable at the minimizer (Theorems 10 and 11). These refute every-start neighborhood guarantees under either regularity regime. They establish no positive-measure failure set and do not refute the almost-sure semialgebraic challenge.

Global and local convergence. We prove global value convergence for centered smooth-plus-sublinear objectives with compact initial sublevels contained in the region where the representation holds, and whole-iterate convergence when the minimizer is unique (Theorem 5). This settles the mixed benchmark above. For coercive finite convex degree-one homogeneous functions, including coercive maxima of finitely many linear functions, every admissible run converges to zero (Theorem 6). Local results cover smooth convex plus finite polyhedral terms and finite polyhedral partitions with smooth extensions (Theorem 7). Moving comparison geometry also yields local convergence for planar smooth maxima under strict-contour and active-gradient nondegeneracy assumptions (Corollary 7). The explicit three-quadratic example 9 has curved switching sets and fails the radial condition, so this extension has a concrete payoff beyond inward radial modeling.

Analysis with degenerating matrices. We prove a hierarchy linking adjacent degrees of the actual principal minors of full-BFGS matrices. Extending the classical trace/determinant potential approach, it forces growth of the arithmetic mean of the reciprocals $1/(g_k^T H_k g_k)$. Comparison energies turn this restriction into convergence: a field must retain information about the objective gap, control the gradient changes it sees, and move with a controlled energy cost. The field is an analytical device, not part of the algorithm. Neither a bounded condition number nor a prescribed spectral limit is needed (Theorems 1–3).

Our conditions are sufficient; they do not classify every successful run. Sections 2 and 3 explain and establish the mechanism; Sections 4 and 5 prove its global and local consequences, and Section 6 gives the deterministic counterexamples. The article includes the essential proofs of these claims. An independent complete companion preserves the wider noncompact and quantitative theory, with Lean proofs, exact correspondence and reproducible audit evidence.

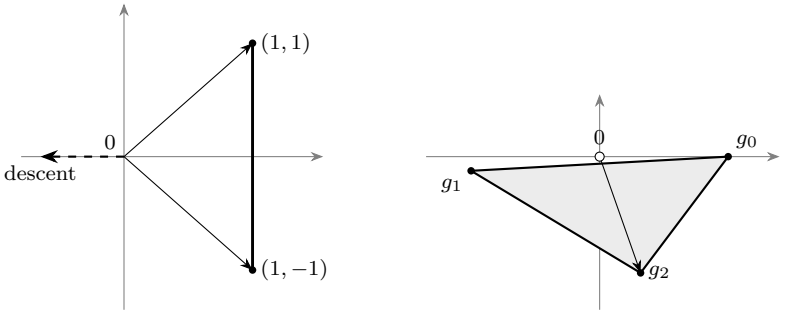
2 What the results say about nonsmooth BFGS

The mixed ridge $F(u, v) = u^2 + |v|$ isolates the difficulty left by a subsequence approaching the nonsmooth line. Reaching $v = 0$ does not eliminate the horizontal error u^2. Yet the vertical gradient jump does not remove the direction

that can reduce this error. Along $v = 0$ the limiting gradients are $(2u, 1)$ and $(2u, -1)$. Their difference is vertical. At a point with $u \neq 0$, the horizontal vector $(-u, 0)$ has the same strictly negative pairing $-2u^2$ with both gradients. The inward radial model also has a small error:

$$F((1-\tau)(u,v)) - F(u,v) + \tau \nabla F(u,v)^T (u,v) = \tau^2 u^2 \qquad (v \neq 0,\ 0 \leq \tau \leq 1).$$

Thus the nonsmooth part is exactly linear under inward scaling, while the smooth part contributes a quadratic remainder. Section 4 uses this property to prove convergence of the whole sequence. The inward segment is a comparison used in the proof; it is not a direction BFGS is assumed to take. No eventual eigenvector or eigenvalue description is needed.



(a) $u^2 + |v|$, $u = 1/2$ (b) Certified nonstationary seed

Fig. 1 Gradient space. Left: the mixed ridge at $u = 1/2$ has a vertical jump and a common horizontal descent direction. Right: the particular three-gradient seed used in Section 6. Its convex hull excludes zero, but its affine hull is the plane. The vertices on the right are rendered from the certified root's rational center; the mathematical definition uses the isolated root. Arrows indicate gradients, not BFGS steps.

The right panel asks a different question: can a genuine full-BFGS run contract around a point where descent still exists? The bad corner is specific. Its slopes are $g_0 = (1, 0)$, $g_1 = (-1, -1/8)$ and $g_2 = (c, d)$, where (c, d) is defined exactly in Appendix C. They are affinely independent, so no nonzero direction pairs equally with all three. Yet $(-1, 16)$ has negative pairing with each, and zero is outside their convex hull. The corner is therefore nonstationary. Section 6 constructs a three-step cycle and verifies that the stated line search produces it, including all rejected trials. This additional dynamical proof is what makes the corner a counterexample.

This distinction between affine and convex balance helps interpret the examples. For a nonempty finite family of slopes a_i, equality $a_i^T v = \kappa < 0$ for every i asks for a common rate of descent, whereas $a_i^T v < 0$ asks only for descent. The former fails when zero belongs to their affine hull; the latter fails when zero belongs to their convex hull. Indeed, a linear functional constant on the slopes is constant on their affine hull. Conversely, if zero is outside that affine hull, its nearest point gives a separating functional constant on the hull;

changing its sign gives the negative common value. The convex version follows from the same closest-point argument for the convex hull. These are elementary separation facts, not a classification of BFGS behavior.

The positive conclusions differ in their geometry and in the locality of the guarantee. Table 1 distinguishes them before the proofs. In every row the quantifier is over infinite admissible runs with arbitrary positive-definite initialization. An admissible run uses actual gradients and satisfies Armijo–weak-Wolfe at its accepted points; it does not assert that an arbitrary line-search implementation always finds such points.

Objective geometry	Starting scope	Conclusion
Centered smooth–sublinear, Theorem 5	Compact initial sublevel in the structural region	Optimal values; unique minimizer gives whole-sequence convergence
Coercive degree-one homogeneous, Theorem 6	Every starting point	Whole sequence converges to zero
Finite polyhedral smooth-piece partition, Theorem 7	Common low sublevel near the unique minimizer	Whole sequence converges to the minimizer
Nondegenerate planar smooth maximum, Corollary 7	Local basin under the stated contour hypotheses	Whole sequence converges to the unique minimizer

Table 1 What the positive theory guarantees. The benchmark $u^2+|v|$ has compact sublevels everywhere, so the first row gives global convergence. The results concern accepted runs, without a presumed matrix limit.

The first two rows make precise global assertions; they do not claim global convergence for every affine maximum. Indeed, the coercive polyhedral realization of the wrong corner in Corollary 8 has a positive limiting objective gap. General affine offsets can place the bad corner away from the true optimizer, whereas homogeneity centers all linear pieces at the origin.

Local behavior raises a further distinction. Finitely many smooth formulas can reproduce bad configurations at infinitely many shrinking scales. A finite polyhedral partition near the optimizer controls the inward model because each remaining cell contains the optimizer after localization. Curved interfaces require a different argument: the planar example in Section 5 has a radial obstruction, yet a moving contour field proves convergence. These are complementary sufficient mechanisms and specific realized failures, rather than a classification by jump span.

3 The analytical framework

Smooth convergence proofs can relate a gradient difference to curvature along a short step. A nonsmooth secant can instead contain a jump that remains large as the step shrinks. Our starting point is to measure how much of that

secant is seen by a direction that still detects the objective gap. A quadratic comparison energy separates the cost of this component from a negative term measuring missed progress. The matrix hierarchy then shows that persistent missed progress cannot be paid for by a bounded secant cost per iteration. The argument uses the matrices actually generated by BFGS throughout.

Definition 1 (Admissible runs) Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, with $n \geq 1$. An admissible run is an infinite sequence of differentiable accepted points x_k, nonzero gradients $g_k = \nabla F(x_k)$, and matrices generated by

$$s_k = -\alpha_k H_k g_k, \qquad x_{k+1} = x_k + s_k, \qquad y_k = g_{k+1} - g_k,$$
$$H_{k+1} = \left(I - \frac{s_k y_k^T}{t_k}\right) H_k \left(I - \frac{y_k s_k^T}{t_k}\right) + \frac{s_k s_k^T}{t_k}, \qquad t_k = y_k^T s_k, \tag{1}$$

where $H_0 \succ 0$, $\alpha_k > 0$, and fixed constants $0 < c_1 < c_2 < 1$ satisfy

$$F(x_{k+1}) \leq F(x_k) + c_1 g_k^T s_k, \qquad g_{k+1}^T s_k \geq c_2 g_k^T s_k. \tag{2}$$

These are the usual actual-gradient, full-BFGS conventions [1]; see also (Xie and Wächter, 2017 preprint, arXiv:1712.08571v1). A zero gradient is successful termination. Statements about admissible runs do not assert that a differentiable acceptable point exists on every ray, or that every implementation finds one.

Write

$$B_k = H_k^{-1}, \qquad q_k = g_k^T H_k g_k, \qquad d_k = -g_k^T s_k = \alpha_k q_k,$$
$$\Delta_k = F(x_k) - F(x_{k+1}), \qquad c = 1 - c_2.$$

Lemma 1 *Every admissible run satisfies $H_k \succ 0$, $q_k, d_k, t_k > 0$ and*

$$c_1 d_k \leq \Delta_k \leq d_k, \qquad t_k \geq c d_k. \tag{3}$$

Moreover,

$$B_{k+1} = B_k - \frac{g_k g_k^T}{q_k} + \frac{y_k y_k^T}{t_k}, \qquad M_k := B_k - \frac{g_k g_k^T}{q_k} \succeq 0, \qquad M_k s_k = 0. \tag{4}$$

If the objective values have a finite limit, then $\sum_k d_k < \infty$.

Proof Given $H_k \succ 0$, nonzero g_k gives $q_k, d_k > 0$; weak Wolfe gives $t_k \geq (1 - c_2) d_k > 0$. For any vector v, the quadratic form in (1) is a sum of two nonnegative terms. If both vanish, then $s_k^T v = 0$ and $(I - y_k s_k^T / t_k) v = 0$, whence $v = 0$. Thus $H_{k+1} \succ 0$, proving the induction. Armijo gives the first lower bound in (3); the supporting inequality for F gives the upper bound. Summing the lower bound proves summability when the values have a finite limit.

The Hessian identity is the classical equivalent form of BFGS [7, Section 5]; it also follows by multiplying its right side by (1), using $B_k s_k = -\alpha_k g_k$ and $s_k^T y_k = t_k$. Finally, metric Cauchy–Schwarz gives $(g_k^T v)^2 \leq q_k v^T B_k v$, and $g_k^T s_k = -\alpha_k q_k$ gives $M_k s_k = 0$.

Lemma 2 (Reference energy) *For any fixed x_*, put $r_k = x_k - x_*$ and*

$$E_k = r_k^T B_k r_k, \qquad a_k = g_k^T r_k, \qquad R_k = \frac{(y_k^T r_{k+1})^2}{t_k}.$$

Then

$$E_{k+1} = E_k - \frac{a_k^2}{q_k} + R_k. \tag{5}$$

If x_* minimizes F, with $F_* = F(x_*)$, then

$$(F(x_N) - F_*)^2 \sum_{k<N} q_k^{-1} \le E_0 + \sum_{k<N} R_k. \tag{6}$$

Proof Since $r_{k+1} = r_k + s_k$ and $M_k s_k = 0$, the Hessian update gives

$$r_{k+1}^T B_{k+1} r_{k+1} = r_k^T M_k r_k + \frac{(y_k^T r_{k+1})^2}{t_k},$$

which is (5). At a minimizer, convexity gives $a_k \ge F(x_k) - F_*$; the values decrease. Summation and $E_N \ge 0$ yield (6).

The identity makes the intended contradiction visible. If a gap of at least $\eta > 0$ persists and the secant correction satisfies $R_k \le C$, then (6) gives

$$\eta^2 \sum_{k<N} q_k^{-1} \le E_0 + CN.$$

Thus an objective-level bound on one component of a gradient jump would force the reciprocal sum to grow at most linearly. The next theorem proves that the full update requires faster growth. Neither step requires a bound on $\|B_k\|$ or a limit for H_k: the initial energy is finite, and the later energies need only remain nonnegative.

3.1 Actual principal minors and reciprocal growth

For a symmetric $n \times n$ matrix H, define

$$e_j(H) = \sum_{\substack{I \subset \{1,\dots,n\} \\ |I| = j}} \det H_{I,I}, \qquad e_{n+1}(H) = 0.$$

These are actual principal determinants. Multilinearity of the determinant identifies them with the coefficients of $\det(I + zH)$; hence they are invariant under orthogonal changes of coordinates and equal the elementary symmetric polynomials of the eigenvalues. In particular $e_1 = \mathrm{tr}$ and $e_n = \det$. Trace and determinant potentials are central to classical BFGS analysis [7,8]. The determinant records volume, while the intermediate principal-minor sums retain information about different-dimensional compressions. The inequality below couples each degree to the next one. After normalization, this coupling supplies a positive weighted potential even when the matrices become poorly conditioned.

Theorem 1 (Principal-minor hierarchy) *Let $H \succ 0$, $s = -\alpha H g$, $\alpha > 0$, and suppose*

$$q = g^T H g > 0, \qquad \beta = -\frac{y^T H g}{q} > 0.$$

For the inverse-BFGS update H_+ in (1)*,*

$$e_j(H_+) \le \frac{\|y\|^2}{\beta^2 q} e_{j+1}(H) + \frac{\alpha}{\beta} e_j(H), \qquad 1 \le j \le n. \tag{7}$$

At $j = n$ the exact identity is $\det H_+ = (\alpha/\beta) \det H$.

Proof Let Z have j orthonormal columns. The complete compression calculation in Lemma 6 gives

$$\det(Z^T H_+ Z) \le \frac{\det \mathrm{Gram}_H(y, Z)}{\beta^2 q} + \frac{\alpha}{\beta} \det(Z^T H Z),$$

where $\mathrm{Gram}_H(y, Z) = [y\ Z]^T H [y\ Z]$. Choose an orthonormal eigenbasis of H, with eigenvalues $\lambda_i > 0$, and write y_i for the coordinates of y in this basis. Summing over its j-column selections Z_I gives

$$\begin{aligned} \sum_{|I|=j} \det \mathrm{Gram}_H(y, Z_I) &= \sum_{|I|=j} \left(\prod_{i \in I} \lambda_i \right) \sum_{\ell \notin I} \lambda_\ell y_\ell^2 \\ &= \sum_{|J|=j+1} \left(\prod_{i \in J} \lambda_i \right) \sum_{\ell \in J} y_\ell^2 \le \|y\|^2 e_{j+1}(H). \end{aligned}$$

The first equality is the Schur complement of the diagonal compression; the second groups terms by $J = I \cup \{\ell\}$. Orthogonal invariance now proves (7). The last assertion follows from the full-size case in Lemma 6.

Theorem 2 (Reciprocal growth) *If an admissible run has bounded gradients and $d_k \to 0$, then*

$$\left(\prod_{k<N} q_k^{-1} \right)^{1/N} \longrightarrow +\infty, \qquad \frac{1}{N} \sum_{k<N} q_k^{-1} \longrightarrow +\infty. \tag{8}$$

Proof Choose $G > 0$ with $\|g_k\| \le G$, and put $K_0 = 4G^2/c^2$. For the actual run,

$$\beta_k = \frac{t_k}{d_k} \ge c, \qquad \|y_k\| \le 2G, \qquad \alpha_k = \frac{d_k}{q_k}.$$

Theorem 1 therefore yields the normalized hierarchy

$$q_k e_j(H_{k+1}) \le K_0 e_{j+1}(H_k) + \frac{d_k}{c} e_j(H_k). \tag{9}$$

This recurrence is derived for the run's principal determinants.

Fix $\eta > 0$ and choose J so that $d_k/c \le \eta$ for $k \ge J$. For $H \succ 0$ define

$$V_\eta(H) = \sum_{j=1}^{n} (K_0/\eta)^{j-1} e_j(H) > 0.$$

Multiply (9) by these weights and sum. The e_{j+1} terms contribute at most $\eta V_\eta(H_k)$, as do the remaining terms. Also $q_k \le G^2 \operatorname{tr} H_k \le G^2 V_\eta(H_k)$. Consequently

$$V_\eta(H_{k+1}) \le \frac{2\eta}{q_k} V_\eta(H_k), \qquad \prod_{k=J}^{J+m-1} q_k^{-1} \ge \frac{1}{G^2 V_\eta(H_J)(2\eta)^{m-1}}.$$

For the last bound, iterate through $m-1$ updates and apply the bound on q_{J+m-1} at the remaining endpoint. Taking mth roots gives a lower limit of at least $(2\eta)^{-1}$. A fixed prefix contributes a factor tending to one. Since η is arbitrary, the geometric mean diverges; AM–GM gives the arithmetic-mean conclusion.

The argument extends the product-estimate tradition in BFGS analysis [6, Theorem 3.7][10, Appendices C–D] to the bounded-gradient, vanishing-decrease regime through the normalized adjacent-degree hierarchy. The assumption $d_k \to 0$ is weaker than the summability supplied by Armijo on a compact sublevel. Finite-horizon bounds and the stronger reciprocal-sum consequence under summability are retained in the companion's Chapter 2; the qualitative conclusion above is sufficient for the article. Neither limit in (8) asserts that $q_k \to 0$ pointwise or that H_k converges.

3.2 Moving comparison directions

The radial direction is effective for a centered homogeneous term, but a curved switching interface may rotate the direction in which its jump cancels. The comparison geometry must then move. Three requirements have distinct roles. *Visibility* keeps the remaining objective gap detectable on a positive band. *Secant control* limits the gradient-jump component seen by the geometry at the next endpoint. *Transport* bounds the cost of changing that geometry between endpoints. The identity $M_k s_k = 0$ means that movement parallel to the executed step carries no cost in the part of the update left after its negative rank-one term. Only the perpendicular projection needs control.

A positive-semidefinite field packages one or several comparison directions. It is constructed from the objective for the analysis; BFGS neither evaluates it nor changes its step in response to it. The rank-one radial field in Section 4 and the contour field in Section 5 illustrate the two designs.

For differentiable x, z, write $g = \nabla F(x)$, $h = \nabla F(z)$, $s = z - x$, $y = h - g$, $d = -g^T s$, $\Delta = F(x) - F(z)$ and $t = y^T s$. Call the pair eligible when

$$d > 0, \qquad c_1 d \le \Delta \le d, \qquad t \ge cd.$$

All accepted pairs are eligible. Requiring a property of every eligible pair is an objective assumption; it does not assert that an arbitrary sequence of eligible pairs can be realized by BFGS. Put $\Pi_s = I - ss^T/(s^Ts)$.

Definition 2 (Band certificate) Let $K = \{F \le b\}$ be compact, with $b > F_* = \min F$. For each nonempty $K_\eta = \{x \in K : F(x) \ge F_* + \eta\}$, $\eta > 0$, a band certificate consists of a finite-valued symmetric PSD field $W_\eta(x)$ on its differentiable points and constants $m_\eta > 0$, $C_\eta \ge 0$, $L_\eta \ge 0$, $\delta_\eta > 0$ such that

$$g^T W_\eta(x) g \ge m_\eta, \tag{10}$$

$$y^T W_\eta(z) y \le C_\eta t, \tag{11}$$

$$\Pi_s W_\eta(z) \Pi_s \preceq (1 + L_\eta d)\Pi_s W_\eta(x) \Pi_s. \tag{12}$$

The first inequality holds at every differentiable point of K_η; the last two hold for all eligible pairs in K_η with $d \le \delta_\eta$.

Neither continuity, uniform boundedness nor positive definiteness of W_η is required. The constants and field may depend on the positive band. Allowing an arbitrary real secant constant gives the same condition, since it can always be replaced by its maximum with zero. The positive-band restriction permits these constants to deteriorate near the solution set: the proof only needs to rule out a limiting gap bounded away from zero. In particular, the condition allows nonsmoothness at an optimizer.

Theorem 3 (Transported comparison certificate) *If K admits the band certificates above, every admissible run with $x_0 \in K$ satisfies*

$$F(x_k) \to F_*, \qquad \operatorname{dist}(x_k, \arg\min F) \to 0.$$

If the minimizer is unique, the iterates converge to it.

Proof Armijo keeps the run in K, and its values have a finite limit. Lemma 1 gives $\sum d_k < \infty$. Gradients are bounded there: choose $\epsilon > 0$ and bound the continuous finite convex function on $K + \epsilon\overline{B}$. Its supporting inequality at x evaluated at $x + \epsilon u$, for every unit u, bounds every subgradient at x. Thus Theorem 2 applies.

Suppose the limiting gap is $\ell > 0$. On a tail $k \ge J$, use the field on $K_{\ell/2}$ and ensure $d_k \le \delta_{\ell/2}$; omit the band subscripts. Set $E_k = \operatorname{tr}(B_k W(x_k)) \ge 0$. Since $M_k = \Pi_{s_k} M_k \Pi_{s_k} \succeq 0$, trace monotonicity and (11)–(12) give

$$\begin{aligned} E_{k+1} &= \operatorname{tr}(M_k W(x_{k+1})) + \frac{y_k^T W(x_{k+1}) y_k}{t_k} \\ &\le (1 + Ld_k)\operatorname{tr}(M_k W(x_k)) + C \le (1 + Ld_k)(E_k - m/q_k) + C. \end{aligned}$$

Trace monotonicity here follows from $\operatorname{tr}(AQ) = \operatorname{tr}(Q^{1/2} A Q^{1/2}) \ge 0$ for PSD A, Q. Let $P_J = 1$ and $P_{k+1} = P_k(1 + Ld_k)$. Summability gives $1 \le P_k \le \exp(L\sum_{j \ge J} d_j) =: \overline{P}$. Dividing by P_{k+1}, summing and using $E_N \ge 0$ yields

$$\frac{m}{\overline{P}} \sum_{k=J}^{N-1} q_k^{-1} \le E_J + C(N - J),$$

contradicting (8). This normalization is the usual summable multiplicative-error argument; compare [11, Lemmas 2.2–2.3]. Finally, a sequence in compact K with values tending to F_* cannot stay a positive distance from the closed solution set: a convergent subsequence would have a nonoptimal limit with value F_*. With a unique minimizer the same argument gives convergence of the entire sequence.

The certificate is a sufficient convergence mechanism. It leaves the metric free to degenerate, and it neither proves finite line-search termination nor requires the iterates to identify a smooth cell. It is also not necessary: the companion's scalar convergence and certificate-nonexistence results (Chapters 5 and 7) separate successful BFGS behavior from existence of this particular comparison field. The planar application below shows a different point: moving geometry can prove convergence where the fixed radial field fails.

4 Global convergence: homogeneous and mixed geometry

The example $F(u, v) = u^2 + |v|$ combines smooth motion along the nonsmooth line with a gradient jump across it. Reaching that line is therefore only part of the convergence question: the smooth coordinate must also approach zero. The comparison energy resolves both motions through a single objective property. Toward the minimizer, a homogeneous term has an exact first-order model; a smooth convex term adds only a quadratic remainder. The argument below works in arbitrary finite dimension and allows the BFGS matrices to become increasingly ill-conditioned.

4.1 Why inward accuracy controls a nonradial algorithm

A gradient jump can make the positive correction in (5) large. We control its pairing with the direction to a minimizer by asking that first-order prediction remain quadratically accurate on short inward segments. This asks nothing about transverse smoothness, and it does not require BFGS to take radial steps. The comparison segments are an analytical device; the actual steps remain those in (1).

Fix a minimizer x_*, write $F_* = F(x_*)$, and let $K = \{x : F(x) \le b\}$ be compact, with $b > F_*$. For differentiable $x, z \in K$, set

$$r_x = x - x_*, \qquad g = \nabla F(x), \qquad h = \nabla F(z),$$
$$T = (h - g)^T (z - x), \qquad N = (h - g)^T r_z.$$

The radial secant condition is the following requirement on positive bands:

$$\begin{gathered}\text{for every } \eta > 0 \text{ there is } C_\eta \ge 0 \text{ such that} \\ N^2 \le C_\eta T \quad \text{whenever } F_* + \eta \le F(z) \le F(x), \quad T > 0.\end{gathered} \tag{13}$$

Here all such differentiable endpoint pairs are included, whether or not they are eligible for the fixed line search. The inward model condition is

$$\begin{gathered}\text{for every } \eta > 0 \text{ there are } M_\eta \geq 0,\ \theta_\eta \in (0, 1/2] \text{ such that} \\ 0 \leq F(x - \tau r_x) - F(x) + \tau \nabla F(x)^T r_x \leq \frac{M_\eta}{2}\tau^2, \quad 0 \leq \tau \leq \theta_\eta,\end{gathered} \tag{14}$$

at every differentiable $x \in K$ with $F(x) \geq F_* + \eta$. The inward point need not be differentiable. It remains in K by convexity. The parameter τ is a fraction of the distance to x_*, rather than a Euclidean step length. Constants may depend on the positive value band: no common bound is required as the band approaches the minimum. Thus nonsmoothness at the solution is permitted, including the absolute-value term in the benchmark.

Theorem 4 (Inward models control radial secants) *For finite convex F and the compact sublevel K above,* (14) *implies* (13).

Proof Choose $G, R > 0$ bounding subgradient norms on K and $\|x - x_*\|$ there. The subgradient bound was proved in Theorem 3; supporting inequalities then give $|F(x) - F(z)| \leq G\|x - z\|$ on K. Use the supporting-plane error

$$D_F(z, x) = F(z) - F(x) - \nabla F(x)^T (z - x) \geq 0.$$

This is the usual Bregman expression [4]; only its supporting endpoint needs to be differentiable. At differentiable endpoints, $T = D_F(z, x) + D_F(x, z)$.

Suppose that (14) holds, and fix a positive band, with constants M, θ. If $N \geq 0$, compare the supporting plane at x with the inward model at $z - \tau r_z$ to obtain

$$\tau N \leq D_F(z, x) + \tfrac{1}{2} M \tau^2.$$

If $N < 0$, reverse the endpoints and use the inward model at x:

$$\tau(T - N) \leq D_F(x, z) + \tfrac{1}{2} M \tau^2.$$

Thus there are $A \in [|N|, 2GR]$ and $D \in \{D_F(z, x), D_F(x, z)\}$ such that $\tau A \leq D + M\tau^2/2$ for $0 \leq \tau \leq \theta$. For $M > 0$ and $A \leq M\theta$, choose $\tau = A/M$, giving $A^2 \leq 2MD$. If $A > M\theta$, take $\tau = \theta$ to obtain $D \geq A\theta/2$ and hence $A^2 \leq (4GR/\theta)D$. When $M = 0$, the same endpoint choice gives an even stronger estimate. Therefore

$$N^2 \leq \max\{2M, 4GR/\theta\}\, T. \tag{15}$$

The converse implication, including inward samples that are not differentiable, is proved in Chapter 4 of the complete companion. The article uses the sufficient direction just proved: objective modeling accuracy controls every descending secant pair on the same positive band.

Corollary 1 (Radial convergence) *Under either condition, every admissible run starting in K converges in value to F_* and in distance to* $\arg\min F$. *With a unique minimizer it converges to x_*.*

Proof Take $W_\eta(x) = r_x r_x^T$. Convexity gives $g^T r_x \geq F(x) - F_* \geq \eta$, proving visibility with $m_\eta = \eta^2$. Equation (13) gives secant control on eligible pairs. Finally $r_z = r_x + s$ implies $\Pi_s r_z = \Pi_s r_x$, so transport holds with $L_\eta = 0$. Apply Theorem 3.

This condition controls only inward first-order modeling error. In a smooth region its infinitesimal expression is $r_x^T \nabla^2 F(x) r_x$, rather than the largest Hessian eigenvalue. A bound almost everywhere on that expression alone would miss a jump of the derivative along a ray; the function-value condition (14) includes such jumps. The condition is sufficient; Theorem 9 will exhibit convergence where it fails on positive bands arbitrarily near the minimizer.

4.2 A family theorem and the mixed benchmark

A finite convex positively homogeneous function is sublinear: its value on $(1-\tau)r$ is exactly $(1-\tau)$ times its value on r. Its angular nonsmoothness consequently disappears from the inward remainder. The next theorem permits an arbitrary locally $C^{1,1}$ convex perturbation, provided the homogeneous part is centered at a minimizer of the sum. Centering is a geometric hypothesis on the objective, not knowledge required by the algorithm. In particular, the two summands need not separately be minimized at that center.

Theorem 5 (Centered smooth-plus-sublinear objectives) *Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, and suppose x_* minimizes F. Let U be an open convex set, let ψ be convex and locally $C^{1,1}$ on U, and let $\gamma : \mathbb{R}^n \to \mathbb{R}$ be finite, convex and positively homogeneous of degree one. Suppose*

$$F(x) = \psi(x) + \gamma(x - x_*) \qquad (x \in U).$$

Every admissible run whose initial sublevel is compact and contained in U satisfies $F(x_k) \to F(x_*)$. The distance to $\arg\min F$ *also tends to zero. If x_* is the unique minimizer, then $x_k \to x_*$. Neither original summand is required to be minimized at x_*.*

Proof Let $K = \{F \leq F(x_0)\}$. Since $g_0 \neq 0$, convexity implies $F(x_0) > F(x_*)$. On compact convex $K \subset U$, local $C^{1,1}$ regularity provides a uniform Lipschitz constant L for $\nabla\psi$: take a finite cover by neighborhoods carrying local constants and subdivide any segment in K into pieces in that cover. Summing the local bounds gives the maximum of those constants along the segment. Integrating the gradient along an inward segment consequently gives

$$\left|\psi(x - \tau r_x) - \psi(x) + \tau \nabla\psi(x)^T r_x\right| \leq \tfrac{1}{2} L \tau^2 \|r_x\|^2.$$

At a differentiable point of F in U, the representation and smoothness of ψ imply differentiability of γ at r_x. Differentiating its homogeneity identity gives $\nabla\gamma(r_x)^T r_x = \gamma(r_x)$. Its contribution to the inward remainder is therefore exactly zero. The upper bound in (14) follows with $M = L \max_{x \in K} \|r_x\|^2$; its lower bound follows from convexity of F. Corollary 1 proves the claim.

The compact initial sublevel supplies uniform constants and traps the accepted iterates. When the representation is global and all sublevels are compact, the theorem applies from every differentiable nonstationary starting point, for every initial SPD matrix and every fixed valid Armijo–weak-Wolfe pair. Neither a lower Hessian bound for the smooth term nor differentiability of the full objective away from the minimizer is required. Without uniqueness, the stated conclusion is value and solution-set distance convergence; convergence to one particular minimizer is not asserted.

Corollary 2 (The quadratic–absolute-value benchmark) *For $F(u, v) = u^2 + |v|$, every infinite admissible full-BFGS run, with arbitrary $H_0 \succ 0$ and any fixed $0 < c_1 < c_2 < 1$, satisfies*

$$F(u_k, v_k) \longrightarrow 0, \qquad (u_k, v_k) \longrightarrow (0, 0).$$

Proof Take $\psi(u, v) = u^2$, $\gamma(u, v) = |v|$, $x_* = (0, 0)$ and $U = \mathbb{R}^2$. The objective is finite and convex, its sublevels are closed and bounded, and its minimizer is unique. More explicitly, for $0 \leq \tau \leq 1$ and $v \neq 0$,

$$F((1 - \tau)(u, v)) - F(u, v) + \tau \nabla F(u, v)^T (u, v) = \tau^2 u^2.$$

Thus the nonsmooth coordinate contributes zero inward error, while the smooth coordinate contributes exactly its quadratic remainder. Theorem 5 applies.

Guo–Lewis proved that every BFGS sequence for this objective has a subsequence approaching a point of the line $v = 0$ [5, Corollary 4.4]; they expressly left convergence of every sequence to the origin open. The corollary above settles that whole-sequence conclusion in the admissible full-BFGS setting. It follows from the family theorem, without a special recurrence for this two-dimensional objective. The conclusion concerns accepted differentiable iterates. It does not assert that every search ray contains a differentiable acceptable point or that a particular search routine can always find one.

4.3 Coercive homogeneous objectives, linear maxima, and gauges

Pure homogeneous geometry is the limiting case with no smooth remainder. Coercivity here has a clear purpose: it prevents escape along a direction of zero or negative homogeneous value and converts convergence in value into convergence to the origin. It imposes no smoothness or finite-piece condition.

Theorem 6 (Coercive homogeneous objectives) *Let $F : \mathbb{R}^n \to \mathbb{R}$, $n \geq 1$, be finite, convex, positively homogeneous of degree one, and coercive. Then zero is its unique minimizer and every infinite admissible full-BFGS run satisfies*

$$F(x_k) \longrightarrow 0, \qquad x_k \longrightarrow 0.$$

The initial point, initial SPD matrix and fixed valid line-search constants are arbitrary within admissibility.

Proof Homogeneity gives $F(0) = 0$. If $F(x) \leq 0$ for some $x \neq 0$, then $F(tx) = tF(x) \leq 0$ as $t \to \infty$, contradicting coercivity. Thus $F(x) > 0$ off zero. Continuity on the unit sphere gives

$$\mu = \min_{\|x\|=1} F(x) > 0, \qquad F(x) \geq \mu\|x\|.$$

All sublevels are compact. Theorem 5, with $\psi = 0$ and $\gamma = F$, gives convergence in value, and the displayed lower bound gives convergence of the iterates. Equivalently, the inward remainder is identically zero, so the radial convergence corollary applies directly.

Corollary 3 (Compact homogeneous models) *The conclusion of Theorem 6 holds for every norm, every finite convex degree-one gauge with bounded unit sublevel, and every finite linear maximum*

$$F(x) = \max_{1 \leq i \leq m} a_i^T x$$

whose zero sublevel is bounded. These are whole-iterate conclusions from every admissible start. The linear maximum has no affine intercepts.

Proof For a norm, continuity and positivity give a positive minimum on the Euclidean unit sphere. For either of the other models, boundedness of the stated sublevel rules out a nonzero x with $F(x) \leq 0$, since its full positive ray would lie in that sublevel. The same sphere argument gives $F(x) \geq \mu\|x\|$, hence coercivity. Finite maxima of linear forms are finite, convex and degree-one homogeneous, so the theorem applies.

No symmetry is needed for the gauge or max-linear cases. The familiar Euclidean norm is an already established consequence of Guo–Lewis's smooth-extension analysis [5, Corollary 4.2]; the scalar absolute-value model was analyzed by Lewis–Overton under their stated search conventions [1, Section 5.1]. The result here treats the stated arbitrary-dimensional homogeneous family. Coercivity is used for this compact, whole-iterate route, not asserted necessary for value convergence. The complete companion, Chapter 7, separately proves the noncompact homogeneous value dichotomy $F(x_k) \to 0$ or $F(x_k) \to -\infty$, together with its bounded-below and additional iterate consequences. Those noncompact conclusions are not needed for, or independently reproved by, the compact global theorems of this article.

5 Local convergence: geometry that survives near a minimizer

The global theorem requires the initial sublevel to lie in the region where its representation is valid. A local theorem must choose that region before seeing a BFGS run or its initial metric. Convexity and a unique minimizer supply precisely this localization. We first use it for finite polyhedral pieces, where inward accuracy has a direct geometric explanation, and then for curved planar switching interfaces, where a moving comparison direction is essential to the proof.

5.1 Localization and finite polyhedral pieces

The distinction between a local objective hypothesis and a hypothesis on the whole initial sublevel is handled by a simple convex trapping argument.

Lemma 3 (A common local basin) *Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex with a unique minimizer x_*. Every sublevel is compact. For each neighborhood U of x_*, there is $\varepsilon > 0$ such that $\{F \le F(x_*) + \varepsilon\} \subset U$. Thus any admissible run starting in that sublevel stays in U, independently of its initial SPD matrix.*

Proof Choose $r > 0$ with $\overline{B}_r(x_*) \subset U$. Continuity, compactness of the sphere and uniqueness give

$$\omega = \min_{\|u\|=r} \big(F(x_* + u) - F(x_*)\big) > 0.$$

For $\|x - x_*\| \ge r$, convexity at the point at distance r on the segment from x_* to x gives

$$F(x) - F(x_*) \ge \frac{\|x - x_*\|}{r}\omega.$$

Consequently F is coercive, hence has compact sublevels, and $\varepsilon = \omega/2$ gives the asserted inclusion. Armijo preserves it.

A finite local partition controls which formulas an inward comparison segment can cross. After removing the finitely many cells that miss x_*, every remaining cell contains x_*. Convexity of a cell then keeps the entire inward segment in that same cell. Its smooth extension yields quadratic model accuracy even if the full gradient jumps across cell boundaries. The algorithm need not identify a cell or a manifold; only the comparison segment uses the cell geometry.

Theorem 7 (Local convergence on finite polyhedral cells) *Let $n \ge 1$ and let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, with a unique minimizer x_*. Suppose an open neighborhood U of x_* is covered by finitely many closed full-dimensional polyhedra P_i. For each i, suppose there is an open set $V_i \supset U \cap P_i$ and a function $f_i \in C^{1,1}_{\mathrm{loc}}(V_i)$ such that*

$$F(x) = f_i(x) \qquad (x \in U \cap P_i).$$

Then there exists $\varepsilon > 0$ such that every infinite full-BFGS run with arbitrary $H_0 \succ 0$, differentiable accepted iterates, fixed Armijo–weak-Wolfe constants $0 < c_1 < c_2 < 1$, and $F(x_0) \le F(x_*) + \varepsilon$ satisfies

$$x_k \in U \quad (k \ge 0), \qquad F(x_k) \to F(x_*), \qquad x_k \to x_*.$$

The basin is chosen before the run and its initial matrix. Only the full objective must be convex; the extensions need not be convex outside their cells.

Proof (Proof of Theorem 7) First discard cells not containing x_*. Each such closed cell has an open complement containing x_*; finiteness supplies one ball that avoids them all. For every remaining cell, its extension domain V_i contains x_*. Shrink to a common ball $B_r(x_*) \subset U \cap \bigcap_{i:x_* \in P_i} V_i$, and then shrink again so that the gradients of all remaining extensions have one Lipschitz bound L there. This uses only finitely many local extensions; they need not have shared a domain before shrinking.

Let x be a differentiability point of F in this ball, and choose a cell P_i containing it. Convexity of P_i places the whole inward segment $x - \tau(x - x_*)$, $0 \le \tau \le 1$, in P_i. Both F and f_i agree along that segment. Comparing their right derivatives at $\tau = 0$ gives

$$\nabla F(x)^T (x - x_*) = \nabla f_i(x)^T (x - x_*).$$

Only equality in the radial direction is needed, including at cell boundaries. The smooth Taylor estimate along the same segment gives

$$0 \le F(x - \tau r_x) - F(x) + \tau \nabla F(x)^T r_x \le \tfrac{1}{2} L \tau^2 \|r_x\|^2 \le \tfrac{1}{2} L r^2 \tau^2.$$

The first inequality uses convexity of F, not convexity of the extension. Lemma 3 supplies a compact positive sublevel wholly contained in this ball, chosen before the run or its matrix. The inward condition holds on that sublevel, so Corollary 1 gives value and iterate convergence. Armijo gives the asserted containment.

Corollary 4 (Smooth losses with polyhedral or quadratic pieces) *A finite convex piecewise linear-quadratic objective with a unique minimizer has the local convergence property of Theorem 7. The same holds for a convex locally $C^{1,1}$ loss plus a finite convex piecewise linear-quadratic regularizer, provided the sum has a unique minimizer.*

Proof By piecewise linear-quadratic we mean a representation by quadratic polynomials on a finite cover by closed full-dimensional polyhedral cells. Each polynomial is a smooth extension of its cell formula; adding the smooth loss preserves the required local $C^{1,1}$ extensions. Theorem 7 applies.

In particular, a finite convex polyhedral penalty, such as a finite sum of absolute-value or hinge terms, is piecewise affine on a finite polyhedral partition. It therefore falls under this corollary when added to a locally $C^{1,1}$ convex loss, under the stated uniqueness assumption. Different cells may have different quadratic Hessians. A finite maximum of arbitrary quadratic formulas need not admit such a polyhedral partition and is not automatically covered. There is also no claim that iterates identify a cell or that rejected trials remain in the local structural neighborhood.

The finiteness and cell geometry are substantive. Finitely many curved switching interfaces need not preserve inward segments, and the negative examples later in the paper show that regularity of smooth branches alone cannot replace the stated geometric hypothesis. The theorem gives a sufficient local basin, without claiming necessity or a uniform rate.

Corollary 5 (A local centered-composite basin) *If the representation and smoothness assumptions of Theorem 5 hold on a neighborhood of a unique minimizer x_*, then some $\varepsilon > 0$, chosen from the objective alone, has the following property: every admissible run starting with $F(x_0) \le F(x_*) + \varepsilon$ stays in that neighborhood and converges to x_* in value and iterates.*

Proof Shrink to an open convex neighborhood where the representation holds. Lemma 3 gives a compact sublevel contained there. Armijo traps the accepted points, and Theorem 5 applies.

For example, a locally $C^{1,1}$ convex loss plus finitely many Euclidean residual norms $\|A_j x - b_j\|$ has this local form. A residual nonzero at x_* remains nonzero nearby, so its Euclidean norm is smooth there; a zero residual gives the centered sublinear function $\|A_j(x - x_*)\|$. Summing these decompositions proves the representation. Under uniqueness, the corollary applies. This example allows nonpolyhedral penalties while retaining the same radial mechanism.

5.2 A moving direction for curved planar switching

The radial direction can fail for a different reason: a curved interface may carry a gradient jump with a nonzero radial component. The secant correction then blows up as endpoints approach opposite sides of the interface. The transported framework allows the comparison direction to follow the interface instead. It must first remain visible to every local supporting model, then cancel the leading jump, and finally move at a cost controlled by the actual objective decrease. The field is constructed from the objective; BFGS never evaluates it.

For $F : \mathbb{R}^2 \to \mathbb{R}$, write

$$\mathcal{C}_{\text{aff}} = \{p : 0 \in \operatorname{aff} \partial F(p)\}.$$

At a point outside this set, let a be the least-norm point of $\operatorname{aff} \partial F(p)$. Orthogonality to its direction space gives $g^T a = \|a\|^2$ for every $g \in \partial F(p)$. Thus $-a/\|a\|^2$ decreases all supporting models at the same unit rate. Affine criticality concerns failure of this common-rate direction; ordinary stationarity concerns failure of every strict common descent direction. They need not coincide.

The two planar hypotheses have separate roles. Strictly convex positive sublevels prohibit long accepted chords with vanishing value decrease. Regular two-branch switching makes each positive-level corner move smoothly with the level and supplies a direction paired equally with both gradients. Neither condition imposes a matrix limit or a lower bound on all eigenvalues of a Hessian. A common positive lower Hessian bound is one sufficient way to obtain strict contours, but is not required.

Theorem 8 (An affine-critical cluster point in the plane) *Let $F : \mathbb{R}^2 \to \mathbb{R}$ be finite and convex, attain its minimum F_*, and have a compact sublevel $K = \{F \le b\}$ with $b > F_*$. Assume the following objective geometry:*

1. *For every $F_* < t \le b$, the boundary of $\{F \le t\}$ contains no nontrivial line segment.*
2. *At every $p \in K \setminus \mathcal{C}_{\mathrm{aff}}$, either F is C^2 in a neighborhood of p, or it equals $\max\{f_1, f_2\}$ there, where f_1, f_2 are convex C^2 functions, both active at p, and $\nabla f_1(p), \nabla f_2(p)$ are linearly independent.*

Every infinite admissible full-BFGS run starting in K has an affine-critical cluster point. No local representation is required at affine-critical points.

Proof The values decrease to $\ell \ge F_*$, the gradients are bounded on K, and $\sum d_k < \infty$. Let Γ be the nonempty compact cluster set. Every point of Γ has value ℓ. If $\ell = F_*$, its points are minimizers and hence affine critical. Suppose, for a contradiction, that Γ contains no affine-critical point. Then $\ell > F_*$.

The steps tend to zero. Otherwise compactness gives a subsequence with $x_k \to x$, $x_{k+1} \to z \ne x$. For $0 \le \theta \le 1$, the supporting inequality and convexity give

$$F(x_k) - \theta d_k \le F((1-\theta)x_k + \theta x_{k+1}) \le (1-\theta)F(x_k) + \theta F(x_{k+1}).$$

Passing to the limit shows $F = \ell$ on $[x, z]$. Such level points lie on the boundary of $\{F \le \ell\}$: if one were interior, extending the line from a minimizer slightly beyond that point and applying convexity would give value strictly less than ℓ. The resulting boundary segment contradicts the first hypothesis.

Lemma 7, proved in Appendix B, constructs a continuous comparison field w on a neighborhood of the whole set Γ. For each sufficiently close eligible pair in that neighborhood, it gives uniform constants $m > 0, C, L \ge 0$ and scalars $\lambda > 0, \beta$ with

$$(g^T w(x))^2 \ge m, \qquad (y^T w(z))^2/t \le C,$$
$$w(z) = \lambda w(x) + \beta s, \qquad \lambda^2 \le 1 + Ld.$$

Compactness implies $\operatorname{dist}(x_k, \Gamma) \to 0$, so these estimates apply to the entire tail, not just a selected subsequence. With $E_k = w(x_k)^T B_k w(x_k)$, the Hessian update and $M_k s_k = 0$ yield

$$E_{k+1} = \lambda_k^2 \left(E_k - \frac{(g_k^T w(x_k))^2}{q_k} \right) + \frac{(y_k^T w(x_{k+1}))^2}{t_k} \le (1 + Ld_k)(E_k - m/q_k) + C.$$

The parenthesized quadratic form is nonnegative. Dividing by $\prod_{j=J}^{k}(1 + Ld_j)$ and summing, exactly as in the proof of Theorem 3, bounds $\sum_{k=J}^{N-1} q_k^{-1}$ by a constant times $N - J + 1$. This contradicts Theorem 2.

Corollary 6 (When affine criticality is correct) *Under the hypotheses of Theorem 8, assume in addition that $\mathcal{C}_{\mathrm{aff}} \cap K \subseteq \arg\min F$. Every admissible run starting in K then converges in value to F_* and in distance to* $\arg\min F$. *A unique minimizer gives convergence of the entire sequence.*

Proof The theorem supplies an affine-critical cluster point, which is optimal by hypothesis. Its value equals the limiting value of the decreasing objective sequence. Hence that value is F_*, and compactness gives the distance and uniqueness conclusions as in Theorem 3.

This condition says that nonoptimal points retain equal-rate descent, which is stronger than ordinary descent. It is a sufficient geometric route in the stated planar setting, not a universal criterion for either convergence or failure. The strict-contour and regular-switching assumptions are independent parts of this route. In particular, the argument neither handles arbitrary flat contours nor transfers automatically to dimension three: the tangent line used in the transport construction is then a higher-dimensional space.

5.3 Nondegenerate smooth maxima and an explicit nonradial basin

The extra affine-correctness condition can be verified from the active models at the minimizer. Positive optimal multipliers ensure that an affine balance cannot lose a branch nearby without losing stationarity; affine independence makes that balance unique and continuous. In the plane these same assumptions leave at most two active branches at a noncritical point, which supplies the required regular switching.

Corollary 7 (Nondegenerate planar smooth maxima) *Let $F : \mathbb{R}^2 \to \mathbb{R}$ be finite and convex with a unique minimizer x_*. Near x_* suppose $F = \max_{1\le i\le m} f_i$, where the branches are convex and C^2. Suppose sufficiently low positive sublevels are strictly convex in the boundary-segment sense of Theorem 8. For the active set I_* at x_*, suppose the gradients are affinely independent and admit weights*

$$\lambda_i^* > 0, \qquad \sum_{i\in I_*} \lambda_i^* = 1, \qquad \sum_{i\in I_*} \lambda_i^* \nabla f_i(x_*) = 0.$$

Then some $\varepsilon > 0$, chosen independently of H_0 and the valid line-search constants, satisfies

$$F(x_0) < F_* + \varepsilon \quad \Longrightarrow \quad F(x_k) \to F_*, \qquad x_k \to x_*$$

for every infinite admissible full-BFGS run. A common positive lower Hessian bound on the locally relevant branches suffices for the contour hypothesis.

Proof Discard the branches inactive at x_* by their strict value gaps. Let $A(x)$ have columns $(\nabla f_i(x), 1)$ for $i \in I_*$. Affine independence makes $A(x_*)$ full column rank, and this persists nearby. If $A(p)\lambda = (0,0,1)^T$ is solvable, its unique solution is

$$\lambda(p) = (A(p)^T A(p))^{-1} A(p)^T (0,0,1)^T.$$

This expression is continuous and equals λ^* at x_*. Shrink the neighborhood until all its coordinates are positive. The subdifferential of a finite smooth maximum is the convex hull of its active gradients. Indeed, differentiating the finite maximum in a direction gives the maximum of the active gradient pairings. Every convex combination of those gradients supports the objective locally; convexity extends the supporting inequality globally by taking a short

initial segment toward any proposed violating point. Conversely, a subgradient outside their convex hull could be strictly separated from that hull by a direction, contradicting the directional-derivative inequality for a subgradient. An affine representation of zero using gradients active at p, extended by zero on inactive indices, must equal $\lambda(p)$. It therefore uses every index and is a convex representation of zero. Thus every nearby affine-critical point is a minimizer.

At most three planar gradients are affinely independent. Away from affine criticality, three cannot be active, since their affine hull would be all of $\mathbb{R}^2$. One active branch gives a smooth chart. For two active branches their gradients are distinct; if they were linearly dependent, their affine line would contain zero. Thus the gradients are independent, and the other branches have strict gaps. This gives the required convex two-branch chart.

Lemma 3 now puts a sufficiently low compact sublevel inside the chosen neighborhood, with its upper level also below the threshold for strict contours. Apply Corollary 6. If all relevant branch Hessians have a common positive lower bound on a convex neighborhood, subtracting the corresponding positive quadratic makes every branch convex. Their maximum is therefore strongly convex there. A sufficiently low sublevel lies in that neighborhood and cannot have a nontrivial boundary segment, proving the last assertion.

The multiplier geometry is related to the nondegeneracy used in local bundle Newton analysis (Lewis and Wylie, 2019 preprint, arXiv:1907.11742v1, Propositions 5.5–5.6) and in Gebken's criticality results [3]. Here the full-BFGS conclusion is derived without assumed iterate or matrix asymptotics. The contour and planar restrictions remain explicit.

Theorem 9 (Convergence with curved, nonradial switching) *For*

$$\begin{aligned} f_0(u,v) &= u + u^2 + \tfrac{1}{2}v^2, \\ f_1(u,v) &= v + \tfrac{1}{2}u^2 + \tfrac{3}{2}v^2, \\ f_2(u,v) &= -u - v + \tfrac{3}{2}u^2 + v^2, \qquad F = \max_i f_i, \end{aligned}$$

every infinite admissible full-BFGS run with $F(x_0) < 1/100$ converges in value and iterates to zero, for arbitrary $H_0 \succ 0$ and fixed valid Armijo–weak-Wolfe constants. Nevertheless, the radial secant condition (13) *fails on a positive band in every neighborhood of zero. The obstructing endpoint pairs can themselves be chosen eligible for $c_1 = 1/4, c_2 = 1/2$.*

Proof The branch Hessians lie between I and $3I$, and $(f_0+f_1+f_2)/3 = u^2+v^2$. Thus $F \geq u^2 + v^2$, zero is the unique minimizer, and all positive contours are strict. The active gradients at zero are $(1,0), (0,1), (-1,-1)$, with positive weights $1/3$. To obtain the stated basin rather than an unspecified one, put $D = 3(1 + v - uv)$. Direct solution of the augmented gradient system gives

$$\begin{aligned} &D\lambda_0 = 1 - 4u + 3v - 7uv, \quad D\lambda_1 = 1 + 2u - 3v - uv, \\ &D\lambda_2 = 1 + 2u + 3v + 5uv. \end{aligned}$$

On $|u|, |v| \le 1/10$, $D \ge 267/100$ and the three numerators are at least $23/100$, $49/100$, and $45/100$. The coefficients sum to one and their weighted gradient is zero. Thus affine independence and affine correctness hold throughout this box, by the same unique-coefficient argument as above. The planar regularity argument also applies throughout the box. Since $\{F \le 1/100\}$ lies there, Corollary 6 proves the exact convergence claim.

For the radial obstruction choose a sufficiently small fixed $u > 0$ and put

$$v = v(u) = \frac{-1 + \sqrt{1 + 4u + 2u^2}}{2} = u - \tfrac{1}{2}u^2 + O(u^3).$$

This is the positive active tie $f_0 = f_1$, with f_2 inactive. At that point

$$(\nabla f_0 - \nabla f_1)^T(u, v) = (1 + u)u - (1 + 2v)v = \tfrac{1}{2}u^2 - v^2 \neq 0$$

for small u, as follows from the displayed expansion. Take $x = (u, v + e)$ and $z = (u, v - e)$ with $e > 0$ tending to zero. Branches 1 and 0, respectively, are uniquely active. A direct calculation gives

$$\begin{aligned} d &= 2e(1 + 3v + 3e), \quad t = 2e(1 + 2v + 4e), \\ \Delta &= e(1 + 4v) + e^2. \end{aligned}$$

As $u, e \to 0^+$, $\Delta/d \to 1/2$ and $t/d \to 1$, so for fixed sufficiently small u and all sufficiently small e, the pairs are eligible for the stated constants. Their radial numerator $y^T z$ tends to $u^2/2 - v^2 \neq 0$ while $t \to 0^+$. The quotient $(y^T z)^2/t$ is unbounded. For each chosen u both endpoint values stay above half the positive tie value; hence one fixed positive band precedes the unbounded sequence. Choosing u arbitrarily small places that band and those pairs in any given neighborhood of zero.

The example identifies the extra freedom supplied by moving geometry. Curvature of the interface produces a radial jump, while the contour-based field cancels it and charges its own motion to the accepted decrease. The proof uses the original coordinates and the unmodified BFGS update; a nonlinear change of variables or a replacement by polyhedral cells would not establish this conclusion.

6 Failure: a contracting mechanism and local realizations

The positive theorems identify geometry that permits convergence. We now show why convexity and regularity of the branch formulas cannot supply an unrestricted deterministic replacement. The first tempting assertion is global: compact sublevels might seem enough to prevent convergence to a wrong corner. The second is local: for one finite maximum of very smooth convex branches, perhaps every sufficiently close start succeeds. We construct an actual failure of each assertion, and then examine what changes when positive lower curvature is imposed.

The mechanism has three parts. A three-step BFGS cycle returns the inverse matrix to a smaller multiple of itself while approaching a nonstationary corner. Strict function and gradient inequalities force the specified line search to produce that cycle. Finally, local copies of the corner accumulate at the true minimizer of one objective. Matching spatial and objective scales keeps the initial matrix fixed across the copies. The distinction between a formal cycle and an implemented run matters at every part of this argument.

6.1 The specified implementation and the local guarantee at issue

At a differentiable point x, let $g = \nabla F(x)$ and $p = -Hg$. The line search starts with lower bracket $\ell = 0$, upper bracket $h = +\infty$, and trial length $\alpha = 1$. It tests Armijo first and weak Wolfe second:

$$F(x + \alpha p) \le F(x) + c_1 \alpha g^T p, \qquad \nabla F(x + \alpha p)^T p \ge c_2 g^T p. \tag{16}$$

An Armijo rejection sets $h = \alpha$; a Wolfe rejection sets $\ell = \alpha$. The next length is $(\ell + h)/2$ when $h < \infty$, and 2ℓ otherwise. If both tests hold, the length is accepted. This is the unit-first bracketing implementation of Lewis–Overton [1, Algorithm 4.6]. All comparisons in the construction are strict, so strict acceptance tests give the same decisions.

Write the full inverse update as

$$\mathcal{B}(H, g, g^+, \alpha) = \left(I - \frac{sy^T}{y^T s}\right) H \left(I - \frac{ys^T}{y^T s}\right) + \frac{ss^T}{y^T s}, \quad s = -\alpha Hg, \quad y = g^+ - g. \tag{17}$$

No resetting or rescaling of the stored matrix is performed.

For this routine, even identity initialization does not give a neighborhood in which every differentiable start succeeds under the following regularity assumptions.

Theorem 10 (Local failure with globally smooth convex branches) *There are four globally convex functions $f_i \in C^{2,1}(\mathbb{R}^2)$ such that*

$$0 \preceq \nabla^2 f_i(x) \preceq 64 I \qquad (i = 0, 1, 2, 3; x \in \mathbb{R}^2)$$

and $F = \max_i f_i$ has compact sublevel sets and a unique nonsmooth minimizer at zero. For this one objective there are fixed starting points $x_0^{(n)} \to 0$, all with $H_0 = I$, such that, for every fixed pair

$$0 < c_1 \le \tfrac{1}{3}, \qquad c_1 < c_2 < 1,$$

the deterministic unit-first bracketing implementation of Section 6 produces infinite full-BFGS runs satisfying

$$x_k^{(n)} \to p_n \neq 0, \qquad F(x_k^{(n)}) \downarrow F(p_n) > F(0) = 0, \qquad 0 \notin \partial F(p_n).$$

Every queried point is differentiable. Each line search takes at most five trials, and its accepted lengths repeat $1, 1/4, 1/16$. The objective, initial points, initial

matrix and complete decision tree are the same throughout the displayed parameter range. The supremum of the norms of all accepted and rejected query points in the nth run tends to zero as $n \to \infty$.

The objective and all starts are chosen before the displayed line-search parameters. The failure set supplied by the theorem is a sequence of starts; it is not a positive-measure basin. There is no uniform positive lower curvature requirement on these branches. Earlier prescribed-orbit constructions for smooth nonconvex objectives [14,15] suggest the value of designing the dynamics first. Other failures concern nonconvex objectives [12]; see also (Diao, 2026 preprint, arXiv:2609.09686v1) for a recent smooth nonconvex planar construction. These results do not supply the convex local claim above. The examples in (Lewis and Overton, *Behavior of BFGS with an Exact Line Search on Nonsmooth Examples*, 2008 manuscript) use exact search or subgradient choices, and the scaled memoryless failures of Asl–Overton [13] concern a different matrix update. Here the essential combination is convex branch regularity, full BFGS with the specified search, a common identity initialization, and every query close to the true minimizer.

6.2 Designing a cycle around a wrong corner

We seek a corner with a strict descent direction that the cycle nevertheless fails to follow to the minimum. Three slopes in the plane provide the needed freedom. Begin with $H_0 = I$ and fix the first two slopes so that the unit step has a simple exact update. The third slope has two free coordinates. We choose them to close the matrix cycle at a scalar multiple ρI and choose the starting point to close the displacement cycle at ρz_0. For the maximum of linear functions, those two identities allow the same geometry to repeat at each smaller scale.

The target lengths $1, 1/4, 1/16$ fit the bisections of the unit-first search. Choosing them in the algebra does not prove that the algorithm accepts them: we must also exclude every earlier candidate and verify both tests at the intended endpoint. The strict trial certificate below supplies exactly this missing dynamical condition.

Let (c, d) be the algebraic pair specified in Appendix C. Define

$$g_0 = (1, 0)^T, \qquad g_1 = (-1, -1/8)^T, \qquad g_2 = (c, d)^T, \qquad G(z) = \max_{0 \le i \le 2} g_i^T z.$$

The useful coarse bounds are $0.318 < c < 0.319$ and $-1.026 < d < -1.025$. Put

$$\rho = \frac{129(c - 8d)}{(16c - 127d - 16)(127c + 16d + 129)}, \qquad 0.060 < \rho < 0.061. \tag{18}$$

Set

$$M_0 = I, \quad M_1 = \begin{pmatrix} 129/256 & -1/16 \\ -1/16 & 1 \end{pmatrix}, \qquad M_2 = \mathcal{B}(M_1, g_1, g_2, 1/4), \tag{19}$$

$$s_0 = (-1, 0)^T, \quad s_1 = (127/1024, 1/64)^T, \qquad s_2 = \rho(1 - c, -d)^T. \tag{20}$$

Phase j	Trial α	Active branch	$10^4 G(z_j + \alpha p_j)$	Decision
0	1	1	$(1218, 1219)$	accept
1	1	0	$(3845, 3846)$	reject
1	1/2	0	$(1364, 1365)$	reject
1	1/4	2	$(723, 724)$	accept
2	1	0	$(6687, 6691)$	reject
2	1/2	0	$(3406, 3408)$	reject
2	1/4	0	$(1765, 1767)$	reject
2	1/8	0	$(944, 946)$	reject
2	1/16	0	$(534, 536)$	accept

Table 2 Rational enclosures and the deterministic trial tree for the algebraic seed. Every rejection is an Armijo rejection. The intervals are enclosures, not rounded approximations asserted to be exact values.

The point is then determined by the displacement closure $s_0 + s_1 + s_2 = -(1-\rho)z_0$; explicitly, let

$$z_0 = \frac{1}{1-\rho}\begin{pmatrix} 897/1024 - \rho(1-c) \\ \rho d - 1/64 \end{pmatrix}, \qquad z_{j+1} = z_j + s_j \quad (j = 0, 1, 2). \tag{21}$$

Lemma 4 (Exact cycle and strict trial decisions) *The matrices in* (19) *are positive definite, and*

$$\mathcal{B}(I, g_0, g_1, 1) = M_1, \qquad \mathcal{B}(M_2, g_2, g_0, 1/16) = \rho I, \qquad z_3 = \rho z_0.$$

At z_j, $j = 0, 1, 2$, the unique active gradient is g_j. With $p_j = -M_j g_j$, the entire line-search query list is Table 2. At every rejection the objective strictly exceeds its current value. At every acceptance,

$$G(z_j) + \tfrac{1}{3} g_j^T s_j - G(z_{j+1}) > 0, \qquad g_{j+1}^T s_j > 0 > g_j^T s_j, \tag{22}$$

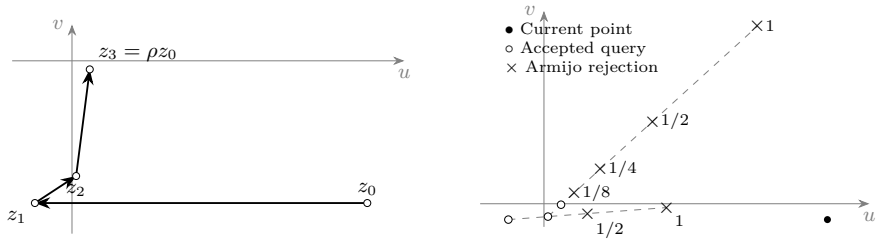
where $g_3 = g_0$. All listed queries and the three starting points have a unique active branch and satisfy

$$|g_i^T z| < 3 (i = 0, 1, 2), \qquad z_2 < 2, \qquad G(z) > 0. \tag{23}$$

Here z_2 in the middle inequality denotes the second coordinate.

The proof of the lemma, including the root isolation, matrix closure and rational certificate, is in Appendix C. The logic of the search can already be read from its margins. Every rejected candidate has a value larger than the current value, so it fails Armijo for every $c_1 > 0$. At each accepted candidate the decrease exceeds one third of the predicted decrease, while the new directional derivative is positive and the old one is negative. Hence Armijo holds for $c_1 \le 1/3$ and weak Wolfe holds for every $c_2 > 0$. Thus the displayed lengths are outputs of the specified search for the entire parameter range in Theorem 10; no accepted-length sequence is assumed. The routine never needs a Wolfe rejection on this example.

Figure 2 displays this unit-first geometry: phase zero accepts its first query, while the next two phases bisect past the rejected candidates before reaching their accepted points. The right panel includes all those rejected candidates.



(a) Accepted cycle; vertical scale enlarged

(b) All queries; rejected lengths labeled

Fig. 2 The accepted three-step seed cycle (left), with closure $z_3 = \rho z_0$, and every rejected candidate point (right). The rendering uses the rational center of the certified root box. It illustrates the certified trial geometry; it is not a proof of exact equality or of membership in the stable manifold used for the perturbed construction.

The finite calculation becomes an infinite run through homogeneity. A smaller copy has the same active slopes, and its inverse matrix produces proportionally smaller steps. Both sides of every line-search comparison have the same scale. The two cycle closures therefore give the exact induction

$$x_{3m+j} = \rho^m z_j, \qquad H_{3m+j} = \rho^m M_j, \qquad j = 0, 1, 2. \tag{24}$$

Each query in phase j is ρ^m times its base query. Gradients are unchanged, function differences and gradient–step products scale by ρ^m, and (17) scales the updated matrix by ρ^m. Thus every branch choice and every rejection is preserved, as well as acceptance. The iterates converge to zero, but the direction $w = (-1, 16)^T$ satisfies

$$g_0^T w = g_1^T w = -1, \qquad g_2^T w = -c + 16d < -1.$$

Hence $G(tw) = -t$ for $t \geq 0$, so this limit is nonstationary.

Corollary 8 (Coercive polyhedral global failure) *The same deterministic run, with $H_0 = I$, fails for*

$$\overline{G}(u, v) = \max\{u, -u - v/8, cu + dv, v - 17\}.$$

This function is coercive and has unique minimizer $(-1, 16)$, with value -1, while the run converges to the nonstationary origin with value zero.

Proof At every query, (23) and $0 < \rho < 1$ give $v - 17 < 0 < G(u, v)$, so the fourth branch is inactive. The convex combination of branches $0, 1, 3$ with weights $8/17, 8/17, 1/17$ is identically -1. Equality in this lower bound forces all three branches to equal -1, hence $(u, v) = (-1, 16)$; the remaining branch is strictly smaller there. If $\overline{G}(u, v) \leq t$, then $u \leq t$, $v \leq t + 17$ and $-u - v/8 \leq t$, which also give lower bounds on both coordinates. Every sublevel is therefore compact.

This already excludes global convergence from every start for coercive finite polyhedral objectives under this routine. Its wrong corner is away from the optimizer, so it is consistent with the local finite-partition convergence theorem. To rule out a local neighborhood guarantee, we need bad corners arbitrarily near the optimizer of the *same* function.

6.3 Matching the spatial scale, objective scale and initial matrix

A change of scale also changes which inverse matrix represents the same search direction. This simple covariance explains both local constructions. Suppose, near the current and queried points under consideration, that

$$F(p + \lambda z) = C + \sigma \mathcal{G}(z), \qquad \lambda, \sigma > 0. \tag{25}$$

Differentiation gives $g_F = (\sigma/\lambda) g_{\mathcal{G}}$. Consequently, for the same candidate length, the corresponding inverse matrix and step are

$$H_F = \frac{\lambda^2}{\sigma} H_{\mathcal{G}}, \qquad p_F = \lambda p_{\mathcal{G}}, \qquad s_F = \lambda s_{\mathcal{G}}. \tag{26}$$

At an accepted pair $y_F = (\sigma/\lambda) y_{\mathcal{G}}$. Thus $y_F^T s_F = \sigma y_{\mathcal{G}}^T s_{\mathcal{G}}$; the two projection factors in the inverse update are identical, and its rank-one term is multiplied by λ^2/σ. This proves the matrix relation in (26) again after the update. Function differences and all gradient–step products are multiplied by $\sigma > 0$, so the Armijo and weak-Wolfe comparisons, including their strict signs, agree.

This is a covariance calculation, not an existence theorem for a local trajectory. To transfer a particular run, the local model must cover the current point and every candidate the search actually visits. Under that containment, induction on the bracket decisions transfers the full trial tree, and induction on accepted steps transfers the matrices. We verify that containment separately in each construction below.

The choice $\sigma = \lambda^2$ is decisive: it leaves the initial matrix unchanged. In the first embedding, the corner's objective height is a_n^3 and its affine slopes are $a_n^2 g_i$. Taking spatial scale $\lambda = a_n^2$ gives $\sigma = a_n^4 = \lambda^2$, so every copy starts with I. Its affine neighborhood has width proportional to a_n, much larger than the entire copied trial tree, whose width is proportional to a_n^2. This separation of scales makes all-query containment possible.

6.4 Convex interpolation that leaves room for every query

We now construct the scalar functions that implement this design. Their node values are a_n^3 and their node slopes are a_n^2. A monotone derivative connects neighboring nodes and is constant on a neighborhood of each node. The constant regions create the local affine corner; the cubic transition makes the second derivative continuous and globally Lipschitz even as the nodes accumulate at zero. The following formulas fix the construction completely.

Let $\theta = 1/16$, $a_n = \theta^n$, $\varepsilon = 1/128$, and $b_* = 2c - 32d \in (33, 34)$. For $b \in \{2, b_*\}$ define

$$t_b = \frac{1 - (1 + \theta + \theta^2)/b}{1 - \theta^2}, \qquad v = \frac{\xi - t_b + \varepsilon}{2\varepsilon}.$$

$$P_b(\xi) = \begin{cases} 0 & \xi \le t_b - \varepsilon, \\ 3v^2 - 2v^3 & |\xi - t_b| < \varepsilon, \\ 1 & \xi \ge t_b + \varepsilon, \end{cases} \qquad J_b(\xi) = \int_0^\xi P_b(t)\, dt.$$

The middle formula for J_b is $2\varepsilon(v^3 - v^4/2)$; its outer formulas are zero and $\xi - t_b$. For $b\theta a_n \le x \le b a_n$, set

$$\varphi_b(x) = (\theta a_n)^3 + b a_n^3 (1 - \theta)\{\theta^2 \xi + (1 - \theta^2) J_b(\xi)\}, \qquad \xi = \frac{x/(b a_n) - \theta}{1 - \theta}. \tag{27}$$

Complete the definition by

$$\varphi_b(x) = \begin{cases} 1 - 3(x+1) + 3(x+1)^2 & x < -1, \\ -x^3 & -1 \le x \le 0, \\ 1 + x - b & x \ge b. \end{cases}$$

Lemma 5 (Scalar interpolation) *The functions φ_b are convex, coercive and globally $C^{2,1}$, are nonnegative, and vanish only at zero. They satisfy*

$$0 \le \varphi_b'' \le \max\{6, 102/b\}, \tag{28}$$

$$\varphi_b(b a_n + h) = a_n^3 + a_n^2 h \qquad (|h| \le a_n/2). \tag{29}$$

Their second derivatives are globally Lipschitz with constant

$$L_b = \max\{6, 139264/(5b^2)\}.$$

Proof The ramp is monotone, C^1, and satisfies $0 \le P_b' \le 96$ and $\int_0^1 P_b = 1 - t_b$. On a positive interval,

$$\varphi_b' = a_n^2\{\theta^2 + (1 - \theta^2) P_b(\xi)\}, \qquad \varphi_b'' = \frac{a_n(1 + \theta)}{b} P_b'(\xi).$$

The identity $\theta^2 + (1 - \theta^2)(1 - t_b) = (1 + \theta + \theta^2)/b$ gives matching values $\varphi_b(b a_n) = a_n^3$. The derivative is constant near both ends of each interval, so the first two derivatives also match, with values a_n^2 and zero at $b a_n$.

As $x \downarrow 0$, the bounds $a_n \le x/(b\theta)$ on its interval give $\varphi_b(x) = O(x^3)$, $\varphi_b'(x) = O(x^2)$ and $\varphi_b''(x) = O(x)$. These match the negative cubic at zero. The formulas also match through second order at -1 and b. The derivative is negative on the negative half-line and positive on the positive half-line; the outer formulas diverge to infinity. This proves convexity, coercivity and the minimum assertions.

The constant-derivative lengths below and above $b a_n$, divided by a_n, are

$$b(1 - \theta)(1 - t_b - \varepsilon), \qquad \frac{b(1 - \theta)(t_b - \varepsilon)}{\theta}.$$

Both exceed $1/2$ for $2 \le b \le 34$, by substitution in the displayed rational formulas; the affine outer extension covers the upper side when $n = 0$. This proves (29). Finally $|P_b''| \le 3/(2\varepsilon^2)$ in each ramp, so differentiation once more gives $|\varphi_b'''| \le 139264/(5b^2)$ there. The negative cubic has third derivative of magnitude six. Continuity of φ_b'' permits summing the Lipschitz estimates over finitely many joins away from zero; taking limits at zero covers the infinitely many accumulating joins and gives the global bound.

6.5 One objective and a uniform induction over all queries

Define

$$F(u, v) = \max\{\varphi_2(u), \varphi_2(-u - v/8), \varphi_{b_*}(cu + dv), v\}. \tag{30}$$

The Hessian of each of the first three branches is a nonnegative scalar multiple of $g_i g_i^T$. Equation (28) gives the upper bounds 51, $51(65/64) < 52$, and 12, respectively, on their operator norms; the last branch has zero Hessian. Their Hessians are globally Lipschitz because φ_b'' is Lipschitz and the arguments are linear. The first two scalar arguments are independent, so their coercivity makes every sublevel of F compact. Nonnegativity and their unique scalar zeros show $F(x) = 0$ exactly at $x = 0$. At zero the first three gradients vanish and the last is $(0, 1)^T$, giving $F'(0; h) = \max\{0, h_2\}$; thus the minimizer is nonsmooth.

Let $p_n = a_n(2, -32)^T$. The first two scalar arguments at p_n equal $2a_n$, and the third equals $b_* a_n$. Consequently

$$F(p_n + h) = a_n^3 + a_n^2 G(h) \quad \text{if } |g_i^T h| \le a_n/2 (i = 0, 1, 2). \tag{31}$$

Indeed, the first three branches are on their affine regions and each has value at least $a_n^3/2$. The first two slab inequalities imply $|h_1| \le a_n/2$ and $|h_2| \le 8a_n$, so the fourth branch is negative. In particular $F(p_n) = a_n^3$ and $F'(p_n; (-1, 16)^T) = -a_n^2 < 0$.

Proof (Proof of Theorem 10) For $n \ge 1$ take $x_0^{(n)} = p_n + a_n^2 z_0$ and $H_0 = I$. We claim that

$$x_{3m+j}^{(n)} = p_n + a_n^2 \rho^m z_j, \qquad H_{3m+j}^{(n)} = \rho^m M_j, \qquad j = 0, 1, 2. \tag{32}$$

At a current point of this form its gradient is $a_n^2 g_j$. Every trial on the finite candidate list therefore has the form $p_n + a_n^2 \rho^m z_{\text{trial}}$. Equation (23) gives

$$|g_i^T (a_n^2 \rho^m z_{\text{trial}})| < 3a_n^2 \le 3a_n/16 < a_n/2.$$

Thus (31) is valid at every candidate query, not only the final accepted one. Unique activity is preserved, and

$$F(p_n + a_n^2 \rho^m z_{\text{trial}}) = a_n^3 + a_n^4 \rho^m G(z_{\text{trial}}), \qquad \nabla F = a_n^2 g_{\text{active}}.$$

These identities instantiate (25) with $\lambda = a_n^2$, $\sigma = a_n^4$ and $C = a_n^3$. The containment just proved applies before each candidate is tested, so the scaling

argument transfers every bracket decision. Equivalently, the additive constant cancels in Armijo and every function difference and gradient–step product is the seed quantity times $a_n^4 \rho^m > 0$. The actual search traverses exactly the candidate list and makes the same strict decisions. In (17), the projection factors are unchanged and the rank-one term has factor ρ^m. The cycle closures prove the next point and matrix identities, completing the induction.

The finite matrices are positive definite. Equation (32) gives convergence to p_n and strict descent to $a_n^3 > 0$. Since the base query set is finite, the norm of every query is bounded by

$$a_n \|(2, -32)\| + a_n^2 \max_{\text{base points and queries}} \|z\|,$$

which tends to zero. The starting points also tend to zero. All other claims were proved directly from (30).

6.6 What changes when positive lower curvature is required

The flat affine regions in the first construction explain why its convex branches have no uniform positive lower curvature. Adding the same positive quadratic to each branch restores that curvature, but changes the dynamics. The old exact cycle can no longer be copied without adjustment. The next result uses a nearby trajectory whose existence is proved analytically, and then matches its scales in a different way.

Theorem 11 (Independent local failure with positive lower curvature) *There is a finite strongly convex function $F : \mathbb{R}^2 \to \mathbb{R}$ with compact sublevels and a unique nonsmooth minimizer at zero such that the following holds. For the fixed deterministic unit-first bracketing implementation of Section 6, with $c_1 = 10^{-4}$, $c_2 = 1/2$ and $H_0 = I$, there are starting points $x_0^{(n)} \to 0$ whose infinite full-BFGS runs converge to nonstationary points $p_n \neq 0$, with $p_n \to 0$ and $F(x_k^{(n)}) \downarrow F(p_n) > F(0)$. Every accepted and rejected query is differentiable, the accepted lengths repeat $1, 1/4, 1/16$, and the supremum of all query norms in the nth run tends to zero.*

The objective can be a maximum of four globally $C^{1,1}$, uniformly strongly convex branches with globally Lipschitz gradients. There are fixed $0 < \underline{\mu} \leq \overline{L} < \infty$ such that

$$\underline{\mu} I \preceq \nabla^2 F(x) \preceq \overline{L} I \quad \text{at every twice-differentiable point.}$$

The first three branches in this construction are not twice differentiable at zero.

Here a classical Hessian is the derivative of the gradient map, with that map defined in a neighborhood of the point. A pointwise second-order expansion alone is not the convention.

The proof in Appendix D has the same sequence of tasks as the first construction. It starts from the explicit strongly convex objective $Q_\mu(z) = \mu\|z\|^2/2 + G(z)$, whose true minimizer is away from the wrong corner. A

Branch regularity	Uniform curvature	At the accumulation point
Four convex $C^{2,1}$ branches	$0 \preceq \nabla^2 f_i \preceq 64I$ globally	All branch Hessians exist and are Lipschitz; no uniform positive lower bound
Four uniformly strongly convex $C^{1,1}$ branches	Positive lower bound and globally Lipschitz branch gradients	The first three branches are not twice differentiable

Table 3 The two independent local failure regimes. Each uses one fixed objective and $H_0 = I$, with bad starts arbitrarily close to its unique minimizer; all actual accepted and rejected queries are differentiable and remain in neighborhoods shrinking to that minimizer. The respective line-search parameter ranges are stated in Theorems 10 and 11.

normalized three-step return map has the original seed as a hyperbolic boundary fixed point. Its stable manifold contains positive scale states, yielding an actual perturbed bad orbit. The seed's strict finite margins ensure that the same accepted and rejected trial lists persist along that orbit. No rounded initial state is asserted to lie on the stable manifold.

For the local embedding the scalar node values are now a_n^2 and the node slopes a_n. The resulting model is $F(p_n + a_n z) = a_n^2(C_* + Q_\mu(z))$, so $\lambda = a_n$, $\sigma = a_n^2$ and again $\lambda^2/\sigma = 1$. We choose one entire perturbed orbit small enough for every query to fit the affine scalar regions. Its one initial SPD matrix is then used in every copy; one fixed affine change makes it I. The faster variation of the scalar derivative retains global $C^{1,1}$ regularity but prevents second differentiability at the accumulation point. This is why the theorem does not supply a finite strongly convex C^2 local counterexample.

6.7 The deterministic assertions excluded

The constructions answer three concrete questions negatively for the specified unit-first rule. Compact sublevels and a finite polyhedral representation do not guarantee convergence from every starting point. Even with one unique minimizer, four globally convex $C^{2,1}$ branches and a uniform upper Hessian bound do not guarantee a neighborhood in which all identity-initialized runs converge, for any fixed pair in Theorem 10's parameter range. Uniform strong convexity and globally Lipschitz branch gradients do not restore that local guarantee in the separate $C^{1,1}$ class at the parameters of Theorem 11.

The simpler quadratic-branch example in the second proof is global: its bad corner is not the minimizer. It therefore does not decide local convergence near the optimizer of a finite maximum of uniformly strongly convex C^2 branches. That local question survives, as do possible guarantees under further objective geometry. The examples also leave almost-sure or generic-start guarantees open: they establish exceptional starts and no positive-measure failure set. They concern the specified weak-Wolfe implementation, and do not prove failure for every line-search routine or for strong Wolfe. Finally, failure of a sufficient comparison condition alone would prove none of these negative conclusions; the actual implemented trajectories do the work here.

7 Synthesis, verification and the remaining questions

The mixed quadratic–absolute-value benchmark now has a whole-sequence answer: every admissible full-BFGS run for $u^2 + |v|$ converges to its unique minimizer. The same centered-family argument gives global convergence for smooth-plus-sublinear objectives whose compact initial sublevels lie in the region of the structural representation, and recovers the known Euclidean-norm result inside the broader coercive homogeneous class. Locally, finite polyhedral geometry controls the inward model error. The planar theorem goes further: a moving comparison field handles curved switches in the three-quadratic example even though its radial secant condition fails. These are objective-based convergence results, with their distinct compactness and geometric assumptions intact.

The counterexamples place a different limit on deterministic guarantees. They exclude a neighborhood in which every admissible start succeeds for the specified unit-first search, both for finite maxima of convex $C^{2,1}$ branches with bounded Hessians and, independently, for the stated uniformly strongly convex $C^{1,1}$ regime. Simpler polyhedral and quadratic-plus-maximum examples establish global failure near a wrong corner; the local embeddings are needed to place bad starts arbitrarily near the objective's own minimizer. The proof follows the actual full update and every line-search query, so the failure is an algorithmic trajectory, not only a prescribed sequence of accepted points.

The common analytical advance is a route around a full spectral description of the BFGS matrices. The principal-minor hierarchy constrains the decrease metric even while the matrices degenerate. A comparison energy then converts those constraints into progress whenever its directions continue to see the remaining objective gap with controlled secant and transport costs. The radial, finite-partition and planar arguments provide different ways of meeting these requirements. Their success does not make the requirements necessary, or turn an affine-critical obstruction into a universal dynamical classification.

These distinctions leave precise questions. The two local negatives do not give a counterexample with finitely many uniformly strongly convex C^2 branches through the minimizer; the planar positive theorem settles a nondegenerate subclass, not that entire class. Nor do exceptional deterministic starts establish failure with positive probability. The accumulating local constructions do not have a finite semialgebraic description. The almost-sure stationarity, all-trial-value rate and spectral-identification assertions in Lewis–Overton's challenge [1, Challenge 7.1] therefore remain outside the conclusions proved here. We also distinguish convergence of an infinite admissible run from a guarantee of indefinite differentiable execution; a zero gradient is successful termination in this paper.

Formal verification and reproducibility. The companion records Lean 4.19.0 proofs with pinned mathlib, exact statement-to-source correspondence, and compiler and transitive-axiom audit logs. The permitted foundational dependencies are propositional extensionality (`propext`), classical choice

(`Classical.choice`) and quotient soundness (`Quot.sound`). The audits reject `sorryAx`, custom mathematical axioms and reachable unsafe or partial proof artifacts. The principal-minor theorem concerns the actual matrices and determinants, and the negative proofs include the search decisions and all-query induction. The rational certificate and the article's proofs can also be read independently of Lean. Formal checking supports these precise claims; it does not establish foundational consistency, publication priority, fidelity of every prose paraphrase, or floating-point robustness. The kernel and execution environment remain trusted, and author-side correspondence reviews are separate from independent human peer review.

AI assistance and author review. The accompanying source tree contains the article, complete companion, figure-generation code, exact certificates and Lean project. Large language models assisted theorem development, proof reconstruction, formalization, literature searches, software and writing. This assistance was substantive; it was not limited to grammar editing. Human authors must review the claims, citations, novelty assessment and disclosure before submission. This author-review edition asserts no author consent, independent human review, affiliations, grants or public archival identifier.

A The BFGS compression calculation

Lemma 6 *Under the assumptions of Theorem 1, let Z have j orthonormal columns, $1 \le j \le n$. Then*

$$\det(Z^T H_+ Z) \le \frac{\det \mathrm{Gram}_H(y, Z)}{\beta^2 q} + \frac{\alpha}{\beta} \det(Z^T H Z).$$

For $j = n$, $\det H_+ = (\alpha/\beta) \det H$.

Proof Put $v = Hg$, $\kappa = g^T H y = -\beta q$ and $P = I - vy^T/\kappa$. Since $y^T s = \alpha\beta q$, the update becomes

$$H_+ = \mathcal{F} + \frac{\alpha}{\beta q} v v^T, \qquad \mathcal{F} = PHP^T.$$

Write

$$A = Z^T H Z \succ 0, \quad a = Z^T H g, \quad b = Z^T H y, \quad r = y^T H y,$$
$$u = a^T A^{-1} a, \quad w = a^T A^{-1} b, \quad z = b^T A^{-1} b.$$

Then

$$Z^T \mathcal{F} Z = A - [a\ b] \begin{pmatrix} -r/\kappa^2 & 1/\kappa \\ 1/\kappa & 0 \end{pmatrix} [a\ b]^T.$$

The identity $\det(I + UV) = \det(I + VU)$ reduces its determinant to a 2×2 determinant. Expanding gives

$$\det(Z^T \mathcal{F} Z) = \frac{\det A}{\kappa^2} \big((\kappa - w)^2 + u(r - z) \big). \tag{33}$$

Schur complements give, independently,

$$\det \mathrm{Gram}_H(y, Z) = \det A(r - z),$$
$$\det \mathrm{Gram}_H(g, y, Z) = \det A\big((q - u)(r - z) - (\kappa - w)^2\big).$$

Both are Gram determinants and hence nonnegative, including when their column families are dependent. Combining these identities yields

$$\det(Z^T \mathcal{F} Z) = \frac{q \det \mathrm{Gram}_H(y, Z) - \det \mathrm{Gram}_H(g, y, Z)}{\beta^2 q^2} \le \frac{\det \mathrm{Gram}_H(y, Z)}{\beta^2 q}.$$

Adding $\rho a a^T$ to the compression changes the coefficient r/κ^2 by ρ. The same 2×2 expansion therefore changes its determinant by exactly $\rho \det A\, u$. No invertibility of $Z^T \mathcal{F} Z$ is used. For $\rho = \alpha/(\beta q)$, the Schur complement of A in $\mathrm{Gram}_H(g, Z)$ gives $0 \le u \le q$, so the increase is at most $(\alpha/\beta) \det A$. This proves the inequality.

When $j = n$, Z is an orthogonal basis and $u = q$, $w = \kappa$, $z = r$. Equation (33) then vanishes, while the rank-one increase is exactly $(\alpha/\beta) \det A$. Orthogonal invariance gives the stated determinant identity.

B Constructing the planar comparison field

This appendix supplies the geometric ingredient used in Theorem 8. The construction applies near a compact regular part of one positive level. It does not assume that the BFGS iterates converge to one point of that level.

Lemma 7 (One field with all endpoint estimates) *Let $F : \mathbb{R}^2 \to \mathbb{R}$ be finite and convex with minimum F_*, and let $C \subset \{F = \ell\}$ be a nonempty compact set, where $\ell > F_*$. Suppose the boundary of $\{F \le \ell\}$ contains no nontrivial segment and every point of C is non-affine-critical and has the smooth or regular convex two-branch structure in Theorem 8. Fix valid Armijo–weak-Wolfe constants. There exist an open neighborhood $U \supset C$, a compact buffer $Q \supset \overline{U}$, a field w defined on a neighborhood of Q, and constants $\eta, \kappa, \delta > 0$, $C_0, C_1, L \ge 0$ such that:*

1. *The field is locally Lipschitz, bounded on Q, and every $g \in \partial F(p)$, $p \in Q$, satisfies $g^T w(p) \ge \eta$. All subgradients at a fixed point have the same pairing with $w(p)$.*
2. *For every eligible pair $x, z \in U$ with $\|s\| < \delta$, its entire segment lies in Q and*

$$|\det(s, w(x + \theta s))| \ge \kappa \|s\| \quad (0 \le \theta \le 1), \qquad (y^T w(z))^2 / t \le C_0.$$

3. *For every such pair there are $\lambda > 0$ and $\beta \in \mathbb{R}$ with*

$$w(z) = \lambda w(x) + \beta s, \qquad |\log \lambda| \le C_1 d, \qquad d \le 1 \implies \lambda^2 \le 1 + Ld.$$

All neighborhoods, constants and the field are chosen before the endpoint pair or any BFGS matrix.

Proof Corner motion. At a nonsmooth point of C, the local branch-value map $x \mapsto (f_1(x), f_2(x))$ has nonsingular derivative. The inverse function theorem gives a C^2 curve $p_j(t)$ characterized by $f_1(p_j(t)) = f_2(p_j(t)) = t$ for t near ℓ. Its C^1 velocity $v_j(t) = p_j'(t)$ satisfies

$$\nabla f_1(p_j(t))^T v_j(t) = \nabla f_2(p_j(t))^T v_j(t) = 1. \tag{34}$$

In particular, this chart has only one nonsmooth point at level ℓ. A smooth point has a neighborhood with no nonsmooth points. A finite cover of C by smaller buffered charts therefore contains only finitely many level-ℓ corners. After shrinking their charts, the neighborhood of C, and the interval of levels, every nearby nonsmooth point lies on one of these finitely many curves. This is a compactness consequence of the charts, rather than an assumed finite bound on future branch changes.

Joining their directions. At a corner, the two active gradients generate its cone of supporting normals. Its unit directions form a compact arc of width less than π; by (34), every such unit normal n has $n^T v_j(t) > 0$. Normal arcs at distinct level-ℓ corners are disjoint: a shared supporting normal would expose both points and their full segment in the boundary of $\{F \le \ell\}$. The strict-contour hypothesis excludes this. The finitely many arcs have positive angular separation.

Choose fixed smooth angular cutoffs $0 \le \chi_j \le 1$ with disjoint supports, equal to one on slightly enlarged normal arcs. Shrink the level interval so that these plateaus contain the whole moving arcs and each support still satisfies $n^T v_j(t) > 0$ uniformly. With $n(\theta) = (\cos\theta, \sin\theta)$ and $T(\theta) = (-\sin\theta, \cos\theta)$, define

$$h(t, \theta) = 1 + \sum_j \chi_j(\theta)\big(n(\theta)^T v_j(t) - 1\big). \tag{35}$$

Because at most one cutoff is nonzero, this is a convex combination of positive quantities. Thus h is bounded and bounded away from zero on the chosen compact level interval. It is C^1 in t, smooth in θ, and its derivatives used below are bounded. No convex-support-function property of h is needed.

At a smooth point, put $n(\theta(x)) = \nabla F(x)/\|\nabla F(x)\|$ and define

$$w(x) = h(F(x), \theta(x))n(\theta(x)) + h_\theta(F(x), \theta(x))T(\theta(x)). \tag{36}$$

All expressions are periodic, so no global choice of angle is required. Near the jth switching curve the relevant cutoff equals one and all others vanish. Consequently $h = n^T v_j$, $h_\theta = T^T v_j$, and

$$w(x) = v_j(F(x)). \tag{37}$$

This holds on both smooth sides in a whole neighborhood of the curve. Use the same formula on the curve itself. It provides a continuous, locally Lipschitz extension across the gradient jump, while the field is C^1 elsewhere. Smaller compact buffers give uniform field and derivative bounds. At a smooth point,

$$\nabla F(x)^T w(x) = \|\nabla F(x)\| h(F(x), \theta(x)) > 0;$$

at a corner every subgradient pairs to one by (34). On sufficiently small buffers of C, these pairings have a uniform positive lower bound. One way to see the needed uniform gradient bound away from zero is the supporting inequality $\|g\|\|x - x_*\| \geq F(x) - F_*$ on the compact buffer, where x_* is any minimizer and the value gap stays positive. This proves the visibility and common-pairing assertions.

The specific angular formula has a crucial derivative cancellation:

$$\partial_\theta(hn + h_\theta T) = (h + h_{\theta\theta})T. \tag{38}$$

Thus at a smooth point the derivative of w maps the tangent line of a level curve into that same line. In the orthonormal basis (T, n),

$$Dw = \begin{pmatrix} a & b \\ 0 & e \end{pmatrix}$$

with bounded coefficients. The formula $hn + h_\theta T$ is classical in normal-angle geometry [17]; see the explicit formula in (Bogosel, 2022 preprint, arXiv:2203.06981v1, Section 2, equation (4)). Here its derivative cancellation is used for a comparison field, without a convexity requirement on h.

Transversality of eligible steps. Choose nested neighborhoods so that $\overline{U}$ lies in the interior of Q, with all preceding bounds valid on a larger buffer. Sufficiently close pairs in U have their segments in Q. If the claimed determinant lower bound failed at arbitrarily short lengths, compactness would give coalescing endpoints tending to $p \in Q$, normalized steps $s/\|s\| \to e$, and segment points at which the normalized determinant tends to zero. Bounded subgradients and their closed graph give endpoint gradient limits $a, b \in \partial F(p)$. Continuity of w makes e parallel to $w(p)$. Positivity of the common pairing and $d > 0$ force $e = -w(p)/\|w(p)\|$. Hence

$$d/\|s\| \longrightarrow -a^T e > 0, \qquad t/\|s\| \longrightarrow (b - a)^T e = 0,$$

contradicting $t \geq (1 - c_2)d$. This proves uniform transversality, including every point of the segment, after reducing δ.

Transport charged to decrease. Write $q(\theta) = x + \theta s$, $\varphi(\theta) = F(q(\theta))$ and $D(\theta) = \det(s, w(q(\theta)))$. Transversality makes D nonzero with constant sign on $[0, 1]$. At a smooth point the basis (T, n) has determinant -1. Writing $s = s_T T + s_n n$, direct calculation yields

$$\det(s, Dw\, s) = s_n\big[(a - e)s_T + bs_n\big]. \tag{39}$$

The numerator has magnitude at most a constant times $|s_n|\|s\|$. Since $|D| \geq \kappa\|s\|$ and $\varphi' = \|\nabla F\|\, s_n$, the quotient is bounded by a constant times $|\varphi'|$. At a corner, (37) gives $w \circ q = v_j \circ \varphi$. The convex function φ has a right derivative at every $\theta < 1$, and the one-sided chain rule gives the same bound there. Therefore, for $H(\theta) = \log|D(\theta)|$,

$$|H'_+(\theta)| \leq A|\varphi'_+(\theta)| \leq A\big(\varphi'_+(\theta) + 2d\big), \qquad 0 \leq \theta < 1.$$

The last inequality uses monotonicity of the convex right derivative and $\varphi'(0) = -d$.

For clarity, the one-sided comparison needed here requires no finite count of switching intersections. If a continuous function u on $[a, b]$ has $u'_+ \le 0$ on $[a, b)$, then $u(b) \le u(a)$. Otherwise choose $\epsilon > 0$ so that $u(b) - \epsilon b > u(a) - \epsilon a$. The continuous function $u(t) - \epsilon t$ has a minimum at some point before b, where its right derivative must be nonnegative, contradicting the strictly negative bound. Apply this fact to both

$$\pm\big(H(\theta) - H(0)\big) - A\big(\varphi(\theta) - \varphi(0) + 2d\theta\big).$$

Both start at zero and have nonpositive right derivative. Since $\varphi(1) \le \varphi(0)$,

$$|H(1) - H(0)| \le 2Ad.$$

Set $\lambda = D(1)/D(0) > 0$. Because $s, w(x)$ are linearly independent, solving in that basis gives $w(z) = \lambda w(x) + \beta s$; the preceding bound is $|\log \lambda| \le 2Ad$. For $0 < d \le 1$, convexity of the exponential on $[0, 1]$ gives $\lambda^2 \le e^{4Ad} \le 1 + (e^{4A} - 1)d$. This establishes the transport constants.

The secant correction. A finite buffered chart cover supplies a common Lipschitz constant for the branch gradients. First suppose the endpoints use the same smooth branch f. On a larger convex chart, with Lipschitz constant $L > 0$, sufficiently close buffered endpoints satisfy

$$\|\nabla f(z) - \nabla f(x)\|^2 \le 2L\big[f(z) - f(x) - \nabla f(x)^T(z - x)\big].$$

Indeed, subtract the tangent plane at x, then use the descent estimate at z for the step $-(\nabla f(z) - \nabla f(x))/L$. This step stays in the larger chart for close endpoints, and convexity makes the resulting function value no smaller than its value at x. This proves the displayed inequality. It is the local convex-gradient estimate underlying cocoercivity [2]; compare (D. Wachsmuth and G. Wachsmuth, 2022 preprint, arXiv:2204.00282v1, Theorem 2.5). Since the branch values and gradients at both endpoints agree with those of F, the bracket is at most t. Boundedness of w then bounds $(y^T w(z))^2/t$ uniformly.

For different endpoint branches i, j in a two-branch chart, the segment contains a tie point p. There $(\nabla f_j(p) - \nabla f_i(p))^T w(p) = 0$. The branch gradients and w are Lipschitz, so

$$|y^T w(z)| \le A_1 \|s\|.$$

Put $a = \nabla f_i(x)$, $b = \nabla f_j(x)$ and $r = \nabla f_j(z) - \nabla f_j(x)$, so $\|r\| \le L\|s\|$. Then

$$a^T s = -d, \qquad b^T s = t - d - r^T s.$$

The two gradient rows are uniformly independent on the smaller chart. Their inverse bound and eligibility imply

$$\|s\| \le A_2(2d + t + L\|s\|^2) \le A_2 \left(1 + \frac{2}{1 - c_2}\right) t + A_2 L \|s\|^2.$$

For sufficiently small $\|s\|$, absorb the last term to obtain $t \geq a_0\|s\|$ with $a_0 > 0$. Consequently $(y^T w(z))^2/t \leq A_1^2\|s\|/a_0$, uniformly bounded. A finite buffered cover ensures every sufficiently close pair lies in one of the preceding charts. Taking the common smaller distance threshold finishes all three assertions for the same field.

C The algebraic seed and its finite rational certificate

This appendix gives the objects and finite inequalities needed for Lemma 4. There are two logically separate tasks. The algebraic equations close the three-step matrix and displacement cycle; the strict inequalities force the specified search to choose that cycle. Solving the equations alone would leave the algorithmic claim unproved. All decimals defining a box are terminating rationals. No rounded point is substituted for a root or an infinite orbit.

C.1 Existence and uniqueness of the defining root

The two unknown coordinates of the third slope are used to enforce the two prescribed dyadic lengths. The matrix calculation in the next subsection first expresses the lengths needed for closure as rational functions $\beta(c,d)$ and $\gamma(c,d)$. Clearing their positive denominators in $\beta = 1/4$ and $\gamma = 1/16$ gives the following two equations. A contraction certificate then selects one exact solution near a rational center.

Put

$$U = 127c + 16d, \qquad V = 16c - 127d - 16, \qquad R = c - 8d,$$
$$K = 8c^2 + 8d^2 + d - 8, \qquad T = c^2 - c + d^2, \qquad Y = (c-1)^2 + d^2,$$

and let

$$P(c,d) = \begin{pmatrix} 264192RK - UV(U+129) \\ 16VTK - RUY \end{pmatrix}. \tag{40}$$

The root is specified by $P(c,d) = 0$ and $\|(c,d) - \bar{r}\|_\infty \leq 10^{-30}$, where

$$\bar{r}_1 = 0.3184814276271186306464513125029070371791220502360 5,$$
$$\bar{r}_2 = -1.0252321196757487254813508700130041428451190931844 4.$$

Lemma 8 (Rational root certificate) *This closed box contains exactly one root. On the box all of $U, V, R, K, T, Y, U + 129$ are positive, and* (18) *holds.*

Proof Let $D = DP(\bar{r})$ and define its inverse by the ordinary two-by-two adjugate formula $A = \operatorname{adj}(D)/\det D$. The following finite rational certificate uses only the coefficients of (40) and the two displayed rationals:

$$\det D \neq 0, \qquad \max_i |(AP(\bar{r}))_i| < 10^{-50}, \qquad \max_i \sum_{2 \leq |\nu| \leq 5} |a_{i,\nu}|\,|\nu| \leq 7000, \tag{41}$$

where, for $\nu = (\nu_1, \nu_2)$,

$$a_{i,\nu} = -\frac{1}{\nu_1!\nu_2!} \sum_{j=1}^{2} A_{ij}\, \partial^{\nu} P_j(\bar{r}).$$

These formulas specify every rational in the certificate: the index set contains the eighteen monomials of degrees two through five. The bounds in (41) follow by substituting the rational center, differentiating the displayed polynomials, and clearing positive denominators in the three comparisons. Thus they are finite integer inequalities, rather than tolerances for a numerically computed root.

For completeness, their analytic consequence follows directly. The centered residual map $N(h) = h - AP(\bar{r} + h)$ has the exact expansion

$$N_i(h) = -(AP(\bar{r}))_i + \sum_{2 \le |\nu| \le 5} a_{i,\nu} h^{\nu};$$

its linear part vanishes because $AD = I$. On $\|h\|_\infty, \|k\|_\infty \le r = 10^{-30}$, telescoping the factors of each monomial gives $|h^\nu - k^\nu| \le |\nu| r \|h - k\|_\infty$, since $r \le 1$ and $|\nu| \ge 2$. Hence N has Lipschitz constant at most $7000r$ and

$$\|N(h)\|_\infty \le 10^{-50} + 7000 r^2 < r.$$

The closed box is complete, so contraction gives a unique fixed point. Since A is invertible, this is exactly a root of P. This is the standard preconditioned-residual principle; compare Rump [16, Theorem 13.3].

The root box lies in $318/1000 < c < 319/1000$ and $-1026/1000 < d < -1025/1000$. Substitution gives positivity of the stated factors. For example,

$$\frac{2397}{100} \le U \le \frac{24113}{1000}, \quad \frac{119263}{1000} \le V \le \frac{59703}{500}, \quad \frac{4259}{500} \le R \le \frac{8527}{1000}.$$

Multiplying positive endpoints in $129R/[V(U + 129)]$ gives $60/1000 < \rho < 61/1000$. In particular all divisions used below are valid.

C.2 The matrix identity

Proof (Proof of the algebraic part of Lemma 4) The first update in (19) is immediate from $s_0 = (-1, 0)^T$ and $g_1 - g_0 = (-2, -1/8)^T$. For arbitrary (c, d) with the positive factors above, put

$$\beta = \frac{66048RK}{UV(U + 129)}, \quad \gamma = \frac{VTK}{RUY}, \quad \widetilde{\rho} = \frac{2064TK}{U(U + 129)Y}.$$

The rational identities obtained from (17) are

$$\mathcal{B}(\mathcal{B}(M_1, g_1, g_2, \beta), g_2, g_0, \gamma) = \widetilde{\rho} I, \qquad -\gamma \mathcal{B}(M_1, g_1, g_2, \beta) g_2 = \widetilde{\rho}(g_0 - g_2). \tag{42}$$

Here is a way to check the identities without guessing matrix entries. The middle update is $C + \beta E$, because its two projection factors are independent of the step length and its rank-one term is linear in that length. Requiring $(C+\beta E)g_2$ to be parallel to $g_0 - g_2$ gives a linear equation for β; substitution of M_1, g_1, g_2 gives the displayed value. For a unit vector $w \perp (g_0-g_2)$, substitution gives $w^T(C + \beta E)w = \widetilde{\rho}$. With the displayed γ, the last step is $\widetilde{\rho}(g_0 - g_2)$. Its projection factors are the orthogonal projector onto $(g_0-g_2)^{\perp}$, and its rank-one term is $\widetilde{\rho}$ times the complementary projector. Their sum is $\widetilde{\rho} I$, proving (42). Each substitution is a rational identity and can equivalently be checked by clearing their nonzero denominators and expanding.

Equations (40) give exactly $\beta = 1/4$ and $\gamma = 1/16$. Moreover,

$$\widetilde{\rho} - \rho = \frac{2064R}{V(U + 129)} \left(\gamma - \frac{1}{16} \right) = 0.$$

This gives the matrix and displacement closures. The formula for z_0 gives $s_0+s_1+s_2 = -(1-\rho)z_0$. The strict inequalities below imply $(g_{j+1}-g_j)^T s_j > 0$ at each phase. Positive definiteness follows inductively from the full update: its two nonnegative quadratic terms cannot both vanish on a nonzero vector when $y^T s > 0$.

C.3 Sound enclosure of every query

The remaining task is finite because one cycle has only nine queries. For each query we must certify its unique active branch, the descent direction at the current iterate, and the comparison that accepts or rejects it. A positive derivative at each accepted query supplies weak Wolfe for every permitted curvature constant. Strict value increase at each rejected query makes Armijo fail, so the routine reaches exactly the next displayed bisection. This explains why the certificate below contains more than values at the three accepted points.

An interval $[a, b]$ encloses the corresponding real quantity. Addition, negation, multiplication and reciprocal are evaluated by

$$\begin{aligned} [a, b] + [c, d] &= [a + c, b + d], \qquad -[a, b] = [-b, -a], \\ [a, b][c, d] &= [\min(ac, ad, bc, bd), \max(ac, ad, bc, bd)], \end{aligned}$$

and $[a, b]^{-1} = [1/b, 1/a]$ when $0 \notin [a, b]$. These inclusion rules follow from order on the reals; induction on a rational expression proves the soundness of every resulting bound. All endpoints used here are rational, so no directed floating-point rounding is needed.

For the query calculation it suffices to use the larger box

$$c \in \frac{[318481, 318482]}{10^6}, \quad d \in \frac{[-1025233, -1025232]}{10^6}, \quad \rho \in \frac{[6019, 6021]}{10^5}. \tag{43}$$

The first two inclusions follow from the root box; substitution in (18), with positive denominator, gives the third. Evaluate z_j from (21) and use

$$p_0 = (-1, 0)^T, \quad p_1 = (127/256, 1/16)^T, \quad p_2 = 16\rho(1 - c, -d)^T.$$

For a row with active index i, set $q = z_j + \alpha p_j$ and evaluate the following rational expressions by the inclusion rules:

$$(g_i - g_k)^T q (k \neq i), \quad -\alpha g_j^T p_j, \quad \begin{cases} g_j^T z_j + \alpha g_j^T p_j/3 - g_i^T q, & \text{acceptance,} \\ g_i^T q - g_j^T z_j, & \text{rejection.} \end{cases}$$

For an acceptance also evaluate $\alpha g_i^T p_j$. Every expression in its row is strictly greater than the following positive rational lower bound, in the order of Table 2:

$$\frac{1}{10^4}(729, 2417, 145, 75, 5964, 2682, 1041, 221, 20). \tag{44}$$

The same interval operations give the value enclosures in that table. For the three base points, the designated active gaps are positive as well. Evaluating $3 \pm g_i^T q$ for every i, $2 - q_2$, and the designated active value $g_i^T q$ gives positive lower bounds for all base points and all nine queries. This proves (23). All inputs, expressions, interval rules and output bounds of this finite certificate are now specified. The strict active gaps also give differentiability on a neighborhood of each query. Together with the matrix identities, these calculations complete the proof of Lemma 4.

D Positive lower curvature through a perturbed cycle

Theorem 11 uses a different regularity balance from Theorem 10. Adding a positive quadratic changes the seed orbit, so exact homogeneity alone no longer supplies a trajectory. We first prove persistence on a stable manifold and then copy one entire perturbed orbit at all scales. The first step yields a global failure from a suitable matrix; the copying step makes bad starts approach the minimizer of one fixed objective; a final fixed change of coordinates makes their common initial matrix the identity. The price of retaining a positive lower curvature bound is the loss of second differentiability of the branches at the accumulation point. Throughout this appendix the line-search constants are $c_1 = 10^{-4}$ and $c_2 = 1/2$.

D.1 A strongly convex global example

A common positive quadratic leaves the active-branch geometry unchanged and makes every sublevel compact. It perturbs the gradients, however, so the homogeneous seed is no longer an exact orbit. Close to the corner, the quadratic gradient is small relative to the fixed slopes. The task is to turn this smallness into an exact infinite trajectory, with every rejection and acceptance retained.

Use the fixed slopes of Appendix C and let

$$Q_\mu(x) = \tfrac{\mu}{2}\|x\|^2 + G(x), \qquad \mu > 0.$$

The vector

$$\bar{g} = \frac{(1,-16)^T}{257} = \frac{129}{257}g_0 + \frac{128}{257}g_1$$

is the least-norm point of $\operatorname{conv}\{g_0, g_1, g_2\}$: $g_0^T\bar{g} = g_1^T\bar{g} = \|\bar{g}\|^2 = 1/257$, whereas $g_2^T\bar{g} > 1/257$. Therefore the unique minimizer is

$$x_* = -\bar{g}/\mu, \qquad \min Q_\mu = -1/(514\mu). \tag{45}$$

At x_* the first two branches tie with distinct gradients, so the minimizer is nonsmooth. At zero the subdifferential is the nonstationary triangle of the original seed.

Proposition 1 (A perturbed bad orbit) *For every $\mu > 0$ and every neighborhood V of zero, some initial state (x_0, H_0), $H_0 \succ 0$, produces a deterministic full-BFGS run on Q_μ converging to zero, with positive objective values decreasing to zero. Its accepted lengths repeat $1, 1/4, 1/16$, and all accepted and rejected query points are differentiable and belong to V.*

The right object for persistence is the three-step return map after dividing out the shrinking scale. The homogeneous cycle becomes a fixed point on the boundary where that scale is zero. We will verify that it is hyperbolic and that its stable manifold enters the positive-scale region. This gives exact perturbed orbits; closeness of a finite floating-point trajectory would not. To prove the proposition, use normalized coordinates

$$x = \tau z, \qquad H = \tau M(b,e), \qquad M(b,e) = \begin{pmatrix} 1 & b \\ b & e \end{pmatrix}, \qquad \tau > 0.$$

The boundary seed is $P_* = (z_0, 0, 1, 0)$ in coordinates (z, b, e, τ). Keep the input τ fixed during three substeps. Starting with $(Z, M) = (z, M(b,e))$, for $j = 0, 1, 2$ perform

$$\begin{aligned} \gamma_j &= g_j + \mu\tau Z, \qquad S = -\alpha_j M\gamma_j, \qquad Z^+ = Z + S, \\ Y &= g_{j+1} - g_j + \mu\tau S, \\ M^+ &= \left(I - \frac{SY^T}{Y^TS}\right) M \left(I - \frac{YS^T}{Y^TS}\right) + \frac{SS^T}{Y^TS}. \end{aligned} \tag{46}$$

with cyclic gradient indices and $\alpha = (1, 1/4, 1/16)$. For the resulting (Z_3, M_3), let $r = (M_3)_{11}$ and define

$$\mathcal{T}_\mu(z, b, e, \tau) = (Z_3/r, (M_3)_{12}/r, (M_3)_{22}/r, \tau r)\,. \tag{47}$$

The seed closure gives $\mathcal{T}_\mu(P_*) = P_*$ and $r(P_*) = \rho$. All denominators are nonzero there. Hence the map is real analytic on an open neighborhood of P_*, using negative τ only to define this analytic extension.

Lemma 9 (Implementation and hyperbolicity) *On a sufficiently small neighborhood of P_*, positive-τ states follow exactly the three query lists of Table 2 under the actual algorithm, and* (47) *is their return map. Its derivative has eigenvalues*

$$\rho^{-1}, \quad \rho^{-1}, \quad \lambda_s, \quad \lambda_u, \quad \rho, \qquad .038 < \lambda_s < .039, \quad 1.564 < \lambda_u < 1.565.$$

The stable eigenspace contains a vector whose τ coordinate is one.

Proof At a query τZ, the value divided by τ is $G(Z) + \mu\tau\|Z\|^2/2$ and the gradient is $g_i + \mu\tau Z$. The common quadratic does not affect which slope is active. All finite branch gaps and normalized search comparisons are continuous in the normalized state. The strict seed margins therefore persist on a common neighborhood. The physical step and secant are exactly the ones in (46) after restoring the scale, proving agreement with the actual update and the entire search tree.

At $\tau = 0$ the matrix return is independent of z. Let J be the derivative at $(b, e) = (0, 1)$ of its normalized shape map. The following rational enclosures specify the spectral certificate:

$$10^{10} J \ \in \begin{pmatrix} [11123986249, 11123986253] & [6459701025, 6459701029] \\ [7515015701, 7515015708] & [4905136882, 4905136890] \end{pmatrix}. \tag{48}$$

They are obtained by differentiating the rational substeps (46) at $\tau = 0$, then applying the interval rules of Appendix C to

$$c \in [318481427627118, 318481427627119]/10^{15},$$
$$d \in [-1025232119675749, -1025232119675748]/10^{15}.$$

Both intervals contain the defining root box. This is a finite rational derivative calculation: for a product use $D(ab) = a\,Db + b\,Da$, and for an inverse use $D(a^{-1}) = -a^{-2}Da$, with denominators bounded away from zero. Substitution of (48) in $p(\lambda) = \lambda^2 - \operatorname{tr}(J)\lambda + \det J$ gives

$$p(.038) > 0, \qquad p(.039) < 0, \qquad p(1.564) < 0, \qquad p(1.565) > 0.$$

These rational sign comparisons locate two distinct real roots and exhaust the degree-two polynomial; no numerical eigenvector is used.

At $\tau = 0$, $Z_3 = z + v(b, e)$ and r is independent of z. The last coordinate of the return is τr. Thus its full derivative has the block form

$$D\mathcal{T}_\mu(P_*) = \begin{pmatrix} \rho^{-1} I_2 & C & \mu a \\ 0 & J & \mu b' \\ 0 & 0 & \rho \end{pmatrix}.$$

The unspecified blocks are finite derivatives of the displayed rational map and do not affect its spectrum. None of the five eigenvalues is zero or has modulus one. Moreover ρ is not an eigenvalue of the upper left four-dimensional block. Assigning last coordinate one and solving the remaining four linear equations gives a ρ-eigenvector. This proves the last assertion and the lemma.

Proof (Proof of Proposition 1) The inverse function theorem makes the return map a local diffeomorphism. The classical local stable-manifold theorem provides a two-dimensional smooth stable manifold through P_*, tangent to the stable eigenspace; (Dyatlov, *Notes on hyperbolic dynamics*, 2018 preprint, arXiv:1805.11660v1) gives an exposition of this principle and its graph transform proof. By the last assertion of Lemma 9, this manifold contains points with arbitrarily small positive τ. Choose such a point whose forward return orbit stays in the implementation neighborhood and tends to P_*. Shrinking the neighborhood keeps $r > 0$ and $M(b, e) \succ 0$, so $\tau^+ = \tau r$ preserves positivity. The actual intermediate matrices remain positive definite by positive curvature.

The physical iterates are $x_{3m} = \tau_m z_m \to 0$. Intermediate points and all queries are τ_m times bounded continuous functions of the normalized state, so they tend to zero too. Their normalized objective values stay positive by the seed's strict positivity. Armijo makes the values decrease to zero, which exceeds (45). The initial stable-manifold point can be taken sufficiently close to P_* that the entire local stable orbit, and hence every physical query, lies in the prescribed V. This proves the proposition. The initial matrix is supplied by the stable-manifold point, not presumed to be identity or obtained by rounded forward shooting.

D.2 An accumulation embedding with four strongly convex branches

The previous proposition places a bad orbit near its wrong corner, whereas the desired local counterexample needs bad starts near the minimizer. We therefore place shrinking copies around centers p_n converging to that minimizer. The spatial factor will be a_n, and the objective factor a_n^2, which preserves the same inverse matrix by (26). To accommodate the common quadratic, the three scalar branch slopes are shifted so that the quadratic cross term restores the original seed gradients inside every copy.

The scalar interpolation must have value a_n^2 and derivative a_n at its nth node, with an affine neighborhood wide enough to contain all seed queries after scaling. A monotone cubic ramp between these plateaux gives a convex function with Lipschitz derivative. The node slopes divided by the node positions stay of order one, so this construction need not, and will not, have a second derivative at zero.

Fix

$$\mu = 1/4096, \qquad c_* = (2, -32)^T, \qquad q_i = g_i - \mu c_*,$$
$$b_i = q_i^T c_*, \qquad r = 1/128, \qquad a_n = r^n.$$

Here $b_0 = b_1 = 1791/1024$ and $1.7 < b_i < 34$. For each of the two distinct values of b_i, set

$$t_b = \frac{1 - (1 + r)/b}{1 - r}, \qquad \varepsilon = 1/1024.$$

Use the same cubic ramp P_b and primitive J_b as in (27), with this center and width. For $ba_{n+1} \le x \le ba_n$ put

$$\psi_b(x) = (ra_n)^2 + ba_n^2(1-r)\{r\xi + (1-r)J_b(\xi)\}, \qquad \xi = \frac{x/(ba_n) - r}{1-r}. \quad (49)$$

Set $\psi_b(0) = 0$, $\psi_b(x) = x^2/2$ for $x < 0$, and $\psi_b(x) = 1 + x - b$ for $x \ge b$.

On each positive interval,

$$\psi_b' = a_n\{r + (1-r)P_b(\xi)\}, \qquad \psi_b'' = P_b'(\xi)/b, \qquad 0 \le P_b' \le 768.$$

The identity $r + (1-r)(1-t_b) = (1+r)/b$ matches node values and derivatives. Their values are a_n^2 and a_n, respectively. The positive derivative is monotone and constant near every node. Its derivative slopes are uniformly bounded, so summing over intervals and taking a limit at zero proves that ψ_b is globally $C^{1,1}$, convex and coercive, nonnegative with unique zero at the origin. The two constant-derivative lengths, divided by a_n, are

$$b(1-r)(1-t_b-\varepsilon), \qquad b(1-r)(t_b-\varepsilon)/r,$$

both greater than $1/2$ for these values of b. Consequently

$$\psi_b(ba_n + h) = a_n^2 + a_n h \qquad (|h| \le a_n/2). \quad (50)$$

At zero, the difference quotients of ψ_b' along ba_n and $(b+1/4)a_n$ equal $1/b$ and $1/(b+1/4)$, respectively. Thus its second derivative at zero does not exist.

Define

$$\begin{aligned} f_i(x) &= \tfrac{\mu}{2}\|x\|^2 + \psi_{b_i}(q_i^T x) \qquad (i = 0, 1, 2), \\ f_3(x) &= \tfrac{\mu}{2}\|x\|^2 + x_2, \qquad F = \max_i f_i. \end{aligned} \quad (51)$$

Each branch is globally μ-strongly convex and $C^{1,1}$. For the first three branches, the gradient-Lipschitz constants are at most $L_i = \mu + \max\{1, 768/b_i\}\|q_i\|^2$. Direct substitution of the rational root bounds gives $L_0 < 439$, $L_1 < 446$ and $L_2 < 27$; hence 512 is a common constant, also covering the fourth branch. Since $F(x) \ge \mu\|x\|^2/2$ with equality zero at the origin, the sublevels are compact and the minimizer is unique. The first three gradients at zero vanish, whereas the fourth is $(0, 1)^T$, so the minimizer is nonsmooth.

At a differentiable point of a finite maximum of L-smooth functions, all active gradients agree. Inactive branches stay inactive locally; taking the maximum of the active upper Taylor bounds therefore gives the local upper Taylor bound with constant L. Applying this with $L = 512$ and combining it with strong convexity proves

$$\mu I \preceq \nabla^2 F(x) \preceq 512 I \quad \text{whenever } F \text{ is twice differentiable.}$$

This does not require a globally Lipschitz gradient for the maximum itself.

D.3 Copying one entire orbit and fixing the initial matrix

Let $p_n = a_n c_*$ and $C_* = 1 + \mu\|c_*\|^2/2$. For y in a fixed sufficiently small neighborhood of zero, (50) yields

$$F(p_n + a_n y) = a_n^2\{C_* + Q_\mu(y)\} \qquad (n \geq 1). \tag{52}$$

Indeed, $|q_i^T y| \leq 1/2$ puts each old branch in its affine scalar region, and expansion of the common quadratic uses $q_i + \mu c_* = g_i$. The fourth branch is inactive there: its residual $a_n(-32 + y_2)$ is negative, while each old scalar residual is at least $a_n^2/2$.

Choose, by Proposition 1, one bad orbit $(\bar{x}_k, \bar{H}_k)$ all of whose queries lie in this neighborhood. For every n, start at $p_n + a_n\bar{x}_0$ with the same matrix $\bar{H}_0$. Induction on the actual query tree gives

$$x_k^{(n)} = p_n + a_n\bar{x}_k, \qquad H_k^{(n)} = \bar{H}_k.$$

This is (25) with spatial factor a_n and objective factor a_n^2. The covariance calculation (26) preserves the matrix, while function differences and directional products acquire the same positive factor. Thus each rejection, acceptance and updated matrix is preserved. Equation (52) applies to every query because the whole chosen seed orbit lies in its domain. Thus $x_k^{(n)} \to p_n$ and $F(x_k^{(n)}) \downarrow a_n^2 C_* > 0$. The three active gradients at p_n are $a_n g_i$, whose convex hull excludes zero. All query norms are at most $a_n(\|c_*\| + \sup_{\text{seed queries}} \|\bar{x}\|) \to 0$.

The seed supplied by the stable manifold has a positive-definite initial matrix $\bar{H}_0$, common to every copy. Identity initialization is obtained by changing coordinates once for the whole objective and the whole sequence of starts, rather than selecting a different objective for each start. Set $A = \bar{H}_0^{1/2}$ and $\widetilde{F}(z) = F(Az)$. The coordinate identities

$$\widetilde{g} = A^T g, \qquad \widetilde{H} = A^{-1} H A^{-T}, \qquad \widetilde{s} = A^{-1} s$$

preserve both scalar line-search tests and (17); see also [1, Section 2]. They give $\widetilde{H}_0 = I$ for this one fixed transformation. Strong-convexity and smoothness constants become $\underline{\mu} = \mu\lambda_{\min}(\bar{H}_0)$ and $\overline{L} = 512\lambda_{\max}(\bar{H}_0)$. All locality and differentiability statements persist under the invertible linear map. This proves Theorem 11.

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